Less Precise, More Reliable

Quantization for Efficient and Robust Road-Scene Perception

Fabio Arnez

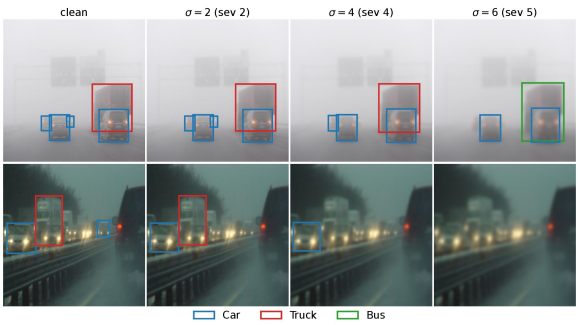
Université Paris-Saclay, CEA, List

IEEE ICVES 2026 · Tutorial, theory session

9 November 2026, Cochabamba, Bolivia

Perception in the vehicle

The sensor relied on most, and the most exposed

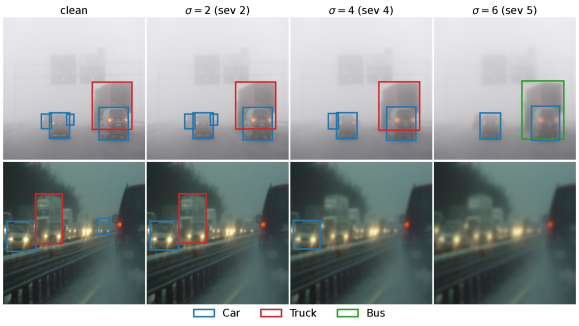


DAWN frames in fog and rain, blurred at three severities; σ is the blur width in pixels, "sev" its severity level. The boxes are a detector's predictions; they thin out and change class as the blur grows (Bouguerra et al., 2026b, Fig. 3).

- Blur, soiling, condensation, rain, and fog act on the lens before any network sees a pixel (Ceccarelli & Secci, 2023).
- Camera perception finds the road objects and names them. This tutorial follows the naming step, a classifier.

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- Camera perception finds the road objects and names them. This tutorial follows the naming step, a classifier.
- A perception failure in such conditions is a safety question. ISO 21448, the safety of the intended functionality (SOTIF), covers hazards caused by performance insufficiencies of sensors and algorithms.

Two demands that pull in opposite directions

A camera-based perception model in a vehicle

Reliable when the image degrades

The model must keep detecting and classifying road objects through blur, weather, and a dirty lens, conditions it may never have seen in training.

Efficient on embedded hardware

Tight limits on cost, power, and heat. Low-bit quantization is the standard answer. W8A8 stores weights and activations as 8-bit integers, a quarter of the 32-bit floating-point (FP32) weight memory, and runs on integer units (Jacob et al., 2018).

The assumed trade-off

Rounding every number to one of 256 levels adds error. A model that already carries error seems a poor candidate to stay accurate when its input degrades too.

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Rounding every number to one of 256 levels adds error. A model that already carries error seems a poor candidate to stay accurate when its input degrades too.

This tutorial asks whether that trade-off **always** holds.

Can less precise be more reliable?

The short answer, which the next 90 minutes earn

- **Yes, in specific and measurable places.** *(measured)*
  The *VLM study* (Bouguerra et al., 2026a) on CLIP-family vision-language models (VLMs) under synthetic corruption, and the *road-scene study* (Bouguerra et al., 2026b) on a road-object classifier under moderate-to-heavy blur, both at W8A8.

Throughout, *measured* means a cited result or a proof on the slide, *plausible* means consistent evidence without a control that isolates the cause, and *untested* means that no experiment has checked it.

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  Most quantized runs still lose accuracy, and the blur gain reverses beyond a window of severities.

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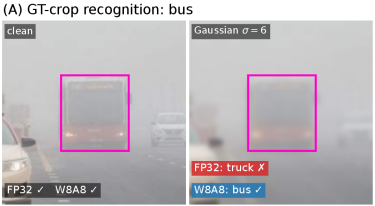
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- **Why, as far as the evidence reaches.**
  Quantization noise and blur perturb the same directions of the features (section 5) *(measured)*; whether the gain needs this is untested.
- **What does the work.** *(untested)*
  Not isolated yet. Clipping set by the learned step (sections 2 and 4) is the leading candidate.

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The running example, a bus in the rain

The road-scene study in one figure



- **(A)** A ground-truth bus from DAWN at Gaussian blur $\sigma = 6$. The FP32 control calls it a truck, the clean-trained W8A8 model a bus, in all three seed pairs.

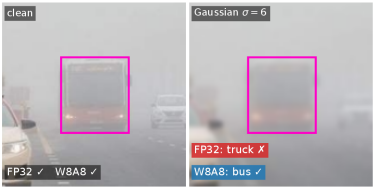
In the figure, a GT crop is the image in one annotated (ground-truth) box, clean-QAT is trained on clean images only, and pp is percentage points. A seed pair is one seed's FP32 control and 8-bit model.

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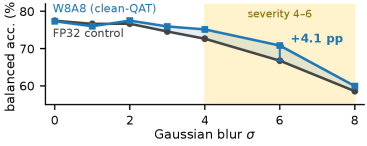
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(A) GT-crop recognition: bus



(B) Road-object recognition under blur



- **(A)** A ground-truth bus from DAWN at Gaussian blur $\sigma = 6$. The FP32 control calls it a truck, the clean-trained W8A8 model a bus, in all three seed pairs.
- **(B)** Over the blur levels, the quantized classifier beats its FP32 control inside a window of severities, $+4.1$ points at $\sigma = 6$. No corrupted image was used in training.

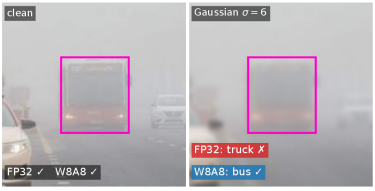
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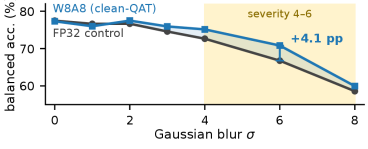
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- **(B)** Over the blur levels, the quantized classifier beats its FP32 control inside a window of severities, $+4.1$ points at $\sigma = 6$. No corrupted image was used in training.
- At the end, the same classifier runs live on a phone, in 8 bits, with half the weight memory of FP16.

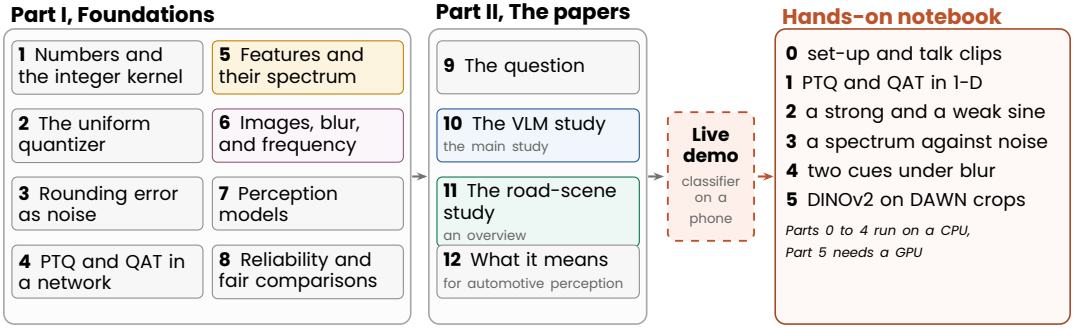
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Section 11 returns to (B), and asks why it happens.

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Roadmap

About 96 minutes of theory, then the hands-on notebook



Questions at the end of the theory session. The companion lecture notes hold the derivations, figures, and numbers with their locators.

What you will be able to do

Learning objectives of the theory session

1. Explain how a quantized perception model is built, range-calibrated, and deployed on integer hardware.
2. Tell the main post-training quantization (PTQ) and quantization-aware training (QAT) methods apart, rotation plus learned step size quantization (Rotation+LSQ) included, and say what each one changes.
3. Measure the reliability of a perception model beyond top-1 accuracy, and compare two models fairly.
4. Place the two featured papers in the efficiency and reliability literatures, for large language models (LLMs), vision models, and automotive perception.
5. Judge when quantization helps robustness, why, and how strong the evidence is.

After the break

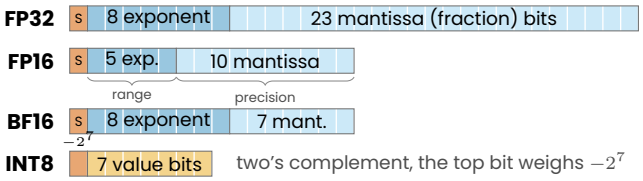
The notebook rebuilds the argument in code, at a scale small enough to manipulate, and re-runs the recipe on the real road-object encoder, where the size of the gain varies with the seed, the run, and the hardware.

1. Numbers and the integer kernel

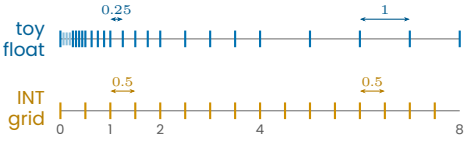
Number formats, one INT8 dot product, fake and true quantization, and speed · Notes §2

Number formats, and what a bit costs

Fewer bits, fewer bytes, and a grid with one gap everywhere



Weight memory. *P* parameters of *b* bits take $M_{\text{w}} = P\,b/8$ bytes. A model of 20.0M parameters takes 80.0 MB in FP32, 40.0 MB in FP16, and 20.0 MB in INT8. W8A8 means 8-bit weights and 8-bit inputs to every quantized layer.



Reading and writing a float

A sign, a power of two, and the digits after a hidden 1

$$x = (-1)^S \, 2^{c-\kappa} \left(1 + \frac{F}{2^m}\right), \qquad \kappa = 2^{n_e - 1} - 1,$$

with the sign bit S, an exponent code c of n_e bits, and a mantissa F of m bits: $1 + 8 + 23$ bits in FP32 and $1 + 5 + 10$ in FP16 (IEEE, 2019), and $1 + 8 + 7$ in BF16, which IEEE 754 does not define.

Decode the FP32 pattern `C0D00000`

1 10000001 1010…0

$S = 1$, so negative. $c = 129$, so $2^{129-127} = 4$.

$1.101_2 = 1 + \frac{1}{2} + \frac{1}{8} = 1.625$.

$x = -4 \cdot 1.625 = -6.5$.

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Encode 0.1, which no float holds exactly

$2^{-4} \le 0.1 < 2^{-3}$, so $c - \kappa = -4$ and $0.1/2^{-4} = 1.6$.
Round $0.6 \cdot 2^m$ to an integer F.

	F	hex	stored value	error
FP32	5033165	`3DCCCCCD`	0.1000000015	$+1.5 \cdot 10^{-9}$
FP16	614	`2E66`	0.0999755859	$-2.4 \cdot 10^{-5}$
BF16	77	`3DCD`	0.1000976563	$+9.8 \cdot 10^{-5}$

Range against precision, and where BF16 fits

Same bytes as FP16, the range of FP32, fewer digits

	bits	largest	smallest normal	gap above 1	digits	integers exact to
FP32	$1+8+23$	$3.40 \cdot 10^{38}$	$1.18 \cdot 10^{-38}$	$1.19 \cdot 10^{-7}$	7.2	$16\,777\,216$
FP16	$1+5+10$	65504	$6.10 \cdot 10^{-5}$	$9.77 \cdot 10^{-4}$	3.3	2048
BF16	$1+8+7$	$3.39 \cdot 10^{38}$	$1.18 \cdot 10^{-38}$	$7.81 \cdot 10^{-3}$	2.4	256

Each entry follows from the formula of the last slide (lecture notes, Table 2).

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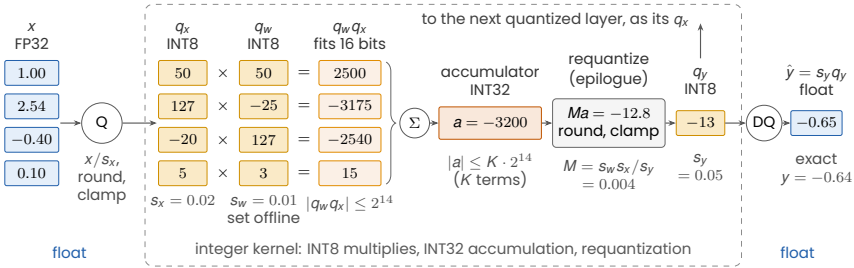
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- **Where.** Mostly to train large networks. With the FP32 range, "no hyper-parameter tuning is required for convergence" (Kalamkar et al., 2019), while FP16 training needs loss scaling and an FP32 copy of the weights (Micikevicius et al., 2018).
- **Benefit.** The FP32 range in half the bytes; FP32 becomes BF16 by rounding away its lower 16 bits.
- **Price.** Eight significant bits. 87.3 is stored as 87.5, and $256+1$ gives 256, so long sums are accumulated in FP32.

Both featured papers deploy INT8 against FP16, not BF16.

One INT8 dot product

Codes and steps, a 32-bit sum, one rounding at the end



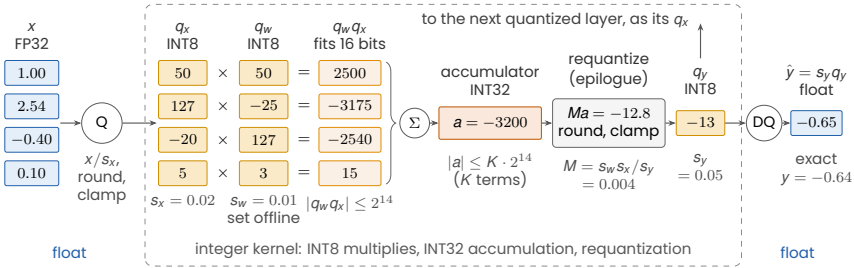
A code and its step

$q = \text{clamp}(\text{round}(v/s), -128, 127)$ codes a number v, so $v \approx sq$. **Max-abs** sets $s = \max|v|/127$, so nothing clips. Weights are coded offline; inputs use a step fixed on a small *calibration set*, coded at run time by a Q node.

A **kernel** is one compiled routine that a processor runs as one unit, here a dot product.

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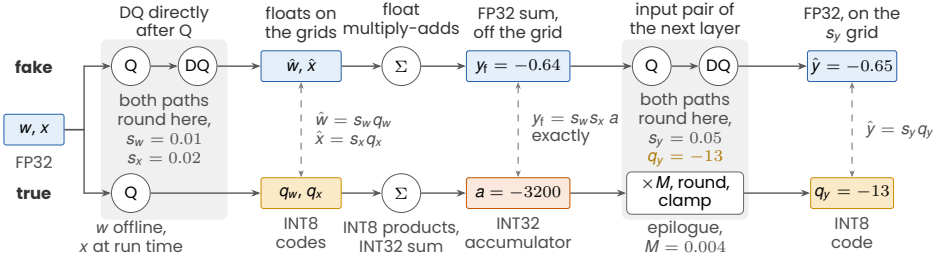
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The steps leave the sum, $y \approx s_w s_x\, a$, and the accumulator $a = \sum_j q_{w,j} q_{x,j}$ is exact integer arithmetic.

Fake and true integer quantization

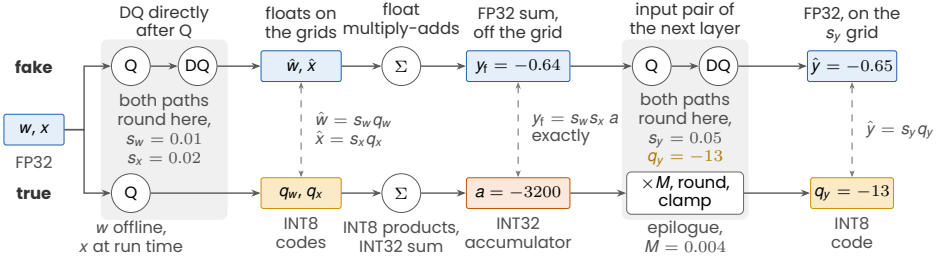
A simulation reproduces the numbers, not the savings



Fake quantization puts the quantizer *Q*, a Q node directly followed by a DQ node, wherever the integer path rounds (Jacob et al., 2018, Sec. 2.4). We simulate to train, which needs gradients, and to run any bit width.

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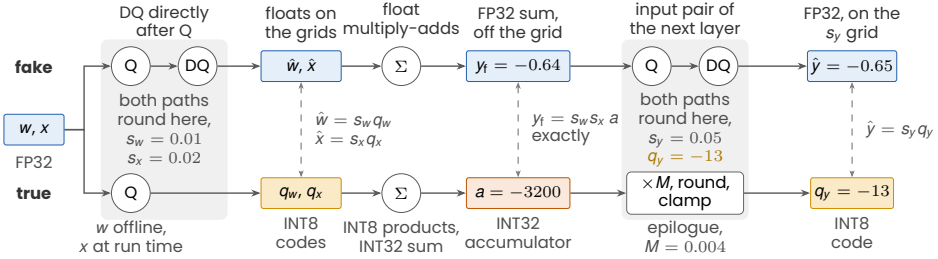


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Deployed effect = simulated effect + drift, the drift being deployed minus simulated accuracy. A simulated gain carries over when the drift is small against it.

Requantization without a float unit

One integer multiply and one shift replace the real factor M

An integer-only processor has no floating-point unit, so it cannot multiply by the real factor $M = s_w s_x / s_y$ of the INT8 figure. It uses a fraction with a power-of-two denominator,

$$M \approx \frac{M_0}{2^n}, \qquad M_0 = \text{round}\left(2^n M\right), \qquad \text{so} \qquad M\,a \approx \frac{M_0\,a}{2^n}.$$

One integer multiplication gives $M_0\,a$, and a right shift by n bits divides it by 2^n, as dropping two decimal digits divides by 100. M_0 and n are fixed before deployment.

The same output, in integers only

$M = 0.004$ and $n = 15$ give $M_0 = \text{round}(131.07) = 131$. Then $-3200 \cdot 131 = -419\,200$, shifted right by 15 bits, is -12.79, which rounds to the same $q_y = -13$.

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Real kernels choose n so that M_0 lies between 2^{30} and 2^{31}, which matches M to about one part in a billion. They round the shift to the nearest integer, for example by adding 2^{n-1} first, and compute $M_0\,a$ in 64 bits (Jacob et al., 2018, Sec. 2.2).

Fake and true quantization can part by one code. For $a = -3126$, fake quantization rounds $Ma = -12.504$ to -13, while the multiplier $131/2^{15}$ gives -12.497 and -12. A 31-bit M_0 makes such cases far rarer, but a network computes millions of outputs.

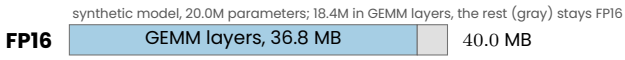
Where the speed comes from

Amdahl's law, and why every speed-up needs its baseline

- Time is arithmetic, bytes moved, and fixed overheads. INT8 helps the first two and can hurt the third, since Q and DQ nodes can break *fusion*, several operations run as one kernel.
- A matrix multiply (GEMM) gains at most $2\times$ over FP16 when its time goes to moving bytes, or the chip's INT8-to-FP16 peak ratio when it goes to arithmetic. Only the part that becomes INT8 speeds up.

Amdahl's law (Amdahl, 1967). A fraction f of the cost sped up by r gives

$$G = \frac{1}{(1-f) + f/r} \leq \frac{1}{1-f}.$$



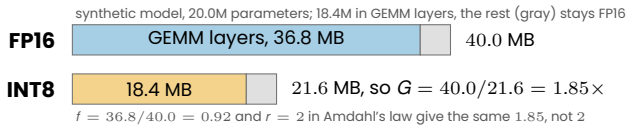
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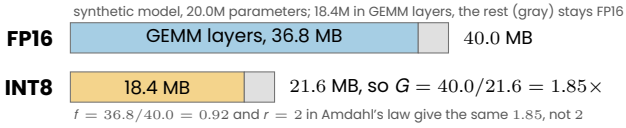
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Amdahl's law, and why every speed-up needs its baseline

- Time is arithmetic, bytes moved, and fixed overheads. INT8 helps the first two and can hurt the third, since Q and DQ nodes can break *fusion*, several operations run as one kernel.
- A matrix multiply (GEMM) gains at most $2\times$ over FP16 when its time goes to moving bytes, or the chip's INT8-to-FP16 peak ratio when it goes to arithmetic. Only the part that becomes INT8 speeds up.

Amdahl's law (Amdahl, 1967). A fraction f of the cost sped up by r gives

$$G = \frac{1}{(1-f) + f/r} \leq \frac{1}{1-f}.$$



A speed-up needs its baseline

Compare INT8 with FP16 on the same stack. In the road-scene study the vendor's INT8 engines ran slower than its own FP16 engine (Bouguerra et al., 2026b).

2. The uniform quantizer

Rounding, clipping, and one knob that sets both · Notes §3

Divide, round, clamp, rescale

Notebook Part 1

The kernel's rule at any bit width, with one real parameter, the step

Symmetric uniform quantizer

$$Q(x) = s \cdot \mathrm{clamp}\big(\mathrm{round}(x/s),\ q_{\min},\ q_{\max}\big),$$

$$\mathrm{clamp}(u, q_{\min}, q_{\max}) = \min\big(\max(u, q_{\min}),\ q_{\max}\big),$$

$$q_{\min} = -2^{b-1}, \qquad q_{\max} = 2^{b-1} - 1.$$

The clamp acts on $u = \mathrm{round}(x/s)$, the code q is stored, and $e(x) = Q(x) - x$ is the error.

Three bits, $s = 0.5$, levels $-2.0, -1.5, \ldots, 1.5$

x	x/s	u	$q = \mathrm{clamp}(u, -4, 3)$	$Q(x)$	$e(x)$

- **A uniform grid.** The levels $sq_{\min}, \ldots, sq_{\max}$ share one gap s, unlike floating point.

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Three bits, $s = 0.5$, levels $-2.0, -1.5, \ldots, 1.5$

x	x/s	u	$q = \mathrm{clamp}(u, -4, 3)$	$Q(x)$	$e(x)$
0.80	1.60	2	2	1.00	+0.20

- **A uniform grid.** The levels $sq_{\min}, \ldots, sq_{\max}$ share one gap s, unlike floating point.

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x	x/s	u	$q = \mathrm{clamp}(u, -4, 3)$	$Q(x)$	$e(x)$
0.80	1.60	2	2	1.00	$+0.20$
-0.20	-0.40	0	0	0.00	$+0.20$

- **A uniform grid.** The levels $sq_{\min}, \ldots, sq_{\max}$ share one gap s, unlike floating point.
- **Mid-tread.** Zero is a level, so every $|x| < s/2$ goes to exactly 0. In this *dead zone* $e = -x$.

Divide, round, clamp, rescale

The kernel's rule at any bit width, with one real parameter, the step

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x	x/s	u	$q = \mathrm{clamp}(u, -4, 3)$	$Q(x)$	$e(x)$
0.80	1.60	2	2	1.00	$+0.20$
-0.20	-0.40	0	0	0.00	$+0.20$
-1.90	-3.80	-4	-4	-2.00	-0.10

- **A uniform grid.** The levels $sq_{\min}, \ldots, sq_{\max}$ share one gap s, unlike floating point.
- **Mid-tread.** Zero is a level, so every $|x| < s/2$ goes to exactly 0. In this *dead zone* $e = -x$.

Divide, round, clamp, rescale

The kernel's rule at any bit width, with one real parameter, the step

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Three bits, $s = 0.5$, levels $-2.0, -1.5, \ldots, 1.5$

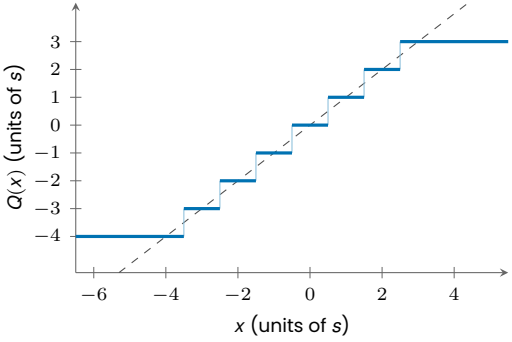
x	x/s	u	$q = \mathrm{clamp}(u, -4, 3)$	$Q(x)$	$e(x)$
0.80	1.60	2	2	1.00	+0.20
−0.20	−0.40	0	0	0.00	+0.20
−1.90	−3.80	−4	−4	−2.00	−0.10
2.30	4.60	5	**3**	1.50	**−0.80**
−2.60	−5.20	−5	**−4**	−2.00	**+0.60**

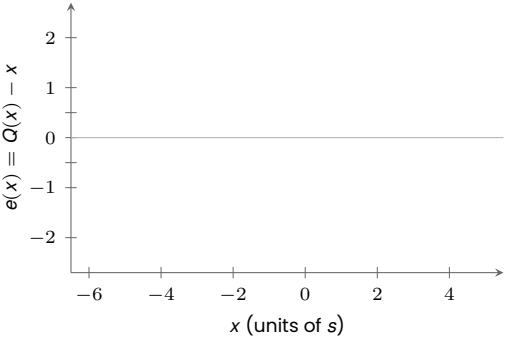
- **A uniform grid.** The levels $sq_{\min}, \ldots, sq_{\max}$ share one gap s, unlike floating point.
- **Mid-tread.** Zero is a level, so every $|x| < s/2$ goes to exactly 0. In this *dead zone* $e = -x$.
- **Clamped.** Outside the clip range the output saturates; the last two rows err by 0.80 and 0.60, past the bound $s/2 = 0.25$.

The staircase and its error

Notebook Part 1

Bounded rounding inside, unbounded clipping outside

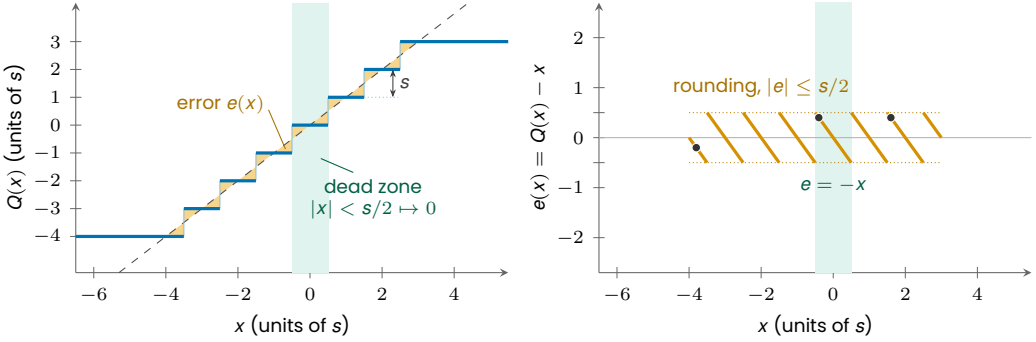




The staircase and its error

Notebook Part 1

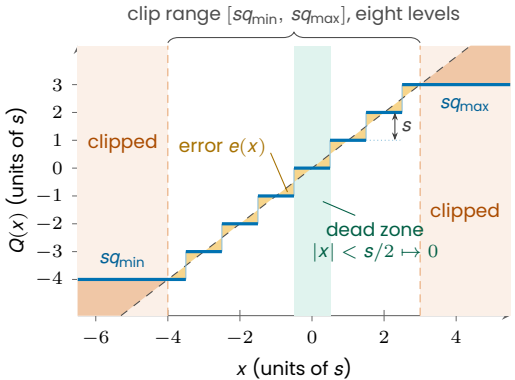
Bounded rounding inside, unbounded clipping outside

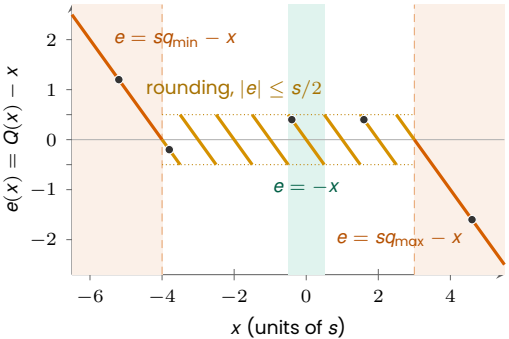


The staircase and its error

Notebook Part 1

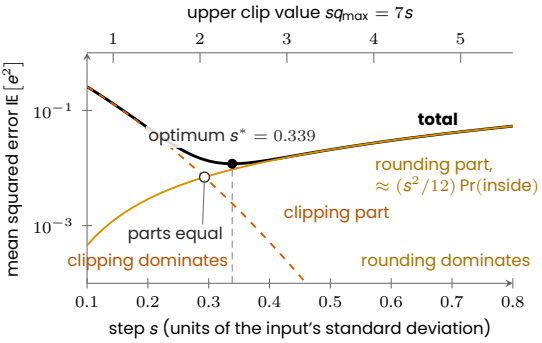
Bounded rounding inside, unbounded clipping outside





The step and the range are one knob

Finer rounding means more clipping, at a fixed bit width

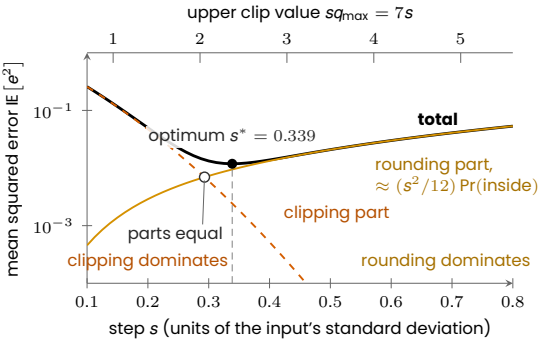


4-bit quantizer, Gaussian input. The optimum clips the 1.2% of inputs outside $[-2.71,\ 2.37]$.

At fixed *b*, halving *s* quarters the rounding variance *and* halves the clip range. In INT8, $s = 0.01$ instead of 0.02 clips $x = 2.0$ to 1.27, an error of -0.73. Choosing *s* is *range calibration*.

The step and the range are one knob

Finer rounding means more clipping, at a fixed bit width



4-bit quantizer, Gaussian input. The optimum clips the 1.2% of inputs outside $[-2.71,\ 2.37]$.

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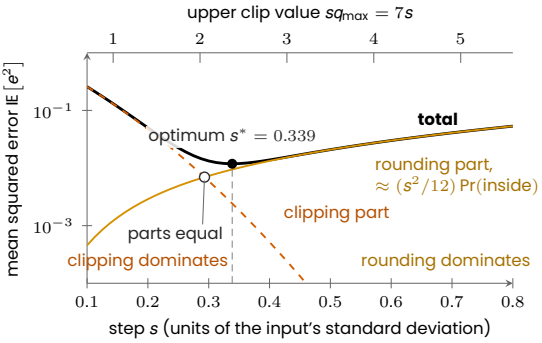
The bus crop in the rain, synthetic

One activation, range-calibrated on clean crops with maximum 2.54, gets $s = 2.54/127 = 0.02$ and the clip range $[-2.56,\ 2.54]$.

A rain streak drives it to 3.10. It is clamped to 2.54, an error of -0.56, or **28 steps**.

The step and the range are one knob

Finer rounding means more clipping, at a fixed bit width



4-bit quantizer, Gaussian input. The optimum clips the 1.2% of inputs outside $[-2.71,\ 2.37]$.

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One activation, range-calibrated on clean crops with maximum 2.54, gets $s = 2.54/127 = 0.02$ and the clip range $[-2.56,\ 2.54]$.

A rain streak drives it to 3.10. It is clamped to 2.54, an error of -0.56, or **28 steps**.

A learned step (LSQ, section 4) is also a learned clip range.

3. Rounding error as noise

Section 2 gave the error of one number. Over many numbers it acts like noise, with exceptions · Notes §4

When rounding error is noise

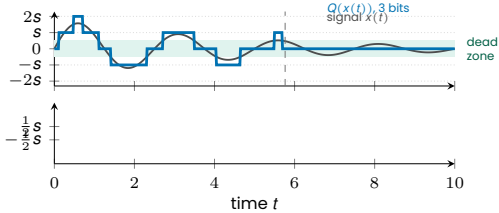
Notebook Parts 1–2

Erased is not the same as buried

The high-resolution model. With a fine step and no clipping, treat $e = Q(x) - x$ as uniform on $[-s/2,\ s/2]$, uncorrelated with the signal and across coordinates. The randomness comes from the input. So

$$\eta^2 = \mathrm{Var}(e) = \frac{s^2}{12}.$$

One more bit halves s and quarters the noise power, so the signal-to-quantization-noise ratio (SQNR) gains about 6 dB per bit at a fixed range. A model, not a law.



3-bit rounding of a decaying signal. The error stays within $\pm s/2$; once the signal stays below $s/2$, $e = -x$.

When rounding error is noise

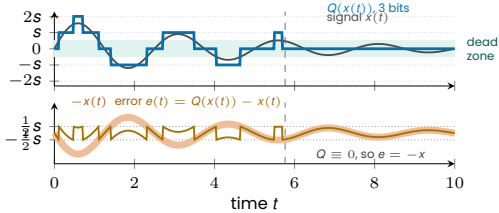
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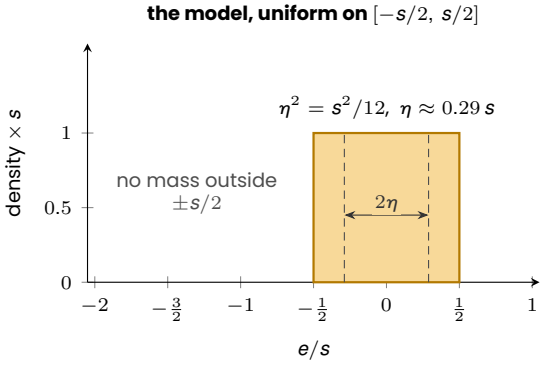


3-bit rounding of a decaying signal. The error stays within $\pm s/2$; once the signal stays below $s/2$, $e = -x$.

- **The dead zone.** Every $|x| < s/2$ goes to 0, so $e = -x$. A weak signal is erased, not drowned. A sine of amplitude 0.15 at $s = 1$ has a model SQNR of -8.7 dB and a true SQNR of 0 dB.
- **Equal inputs, equal errors.** Three coordinates at 0.31 with $s = 0.2$ all become 0.4, each with the error $+0.09$.
- **Clipping.** An unbounded error that opposes the signal.

Quantization noise as a random vector

From one rounding error to an error in every direction



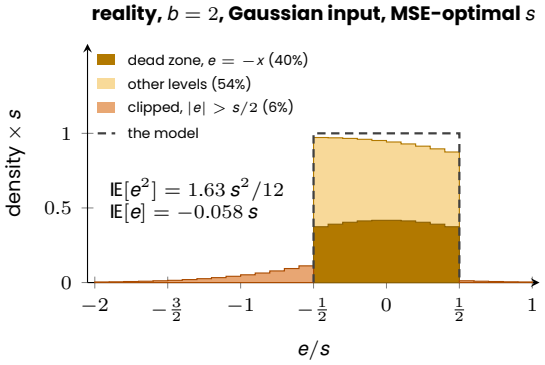
One step for all d coordinates. Under the high-resolution model the errors e_i are independent and uniform on $[-s/2,\ s/2]$, so

$$\mathbb{E}[\boldsymbol{e}] = \mathbf{0}, \quad \mathrm{Cov}(\boldsymbol{e}) = \eta^2 I, \quad \eta^2 = \frac{s^2}{12}.$$

Along any unit direction $\boldsymbol{v}$, $\mathrm{Var}(\boldsymbol{v}^\top \boldsymbol{e}) = \eta^2$, the same power in every direction.

Quantization noise as a random vector

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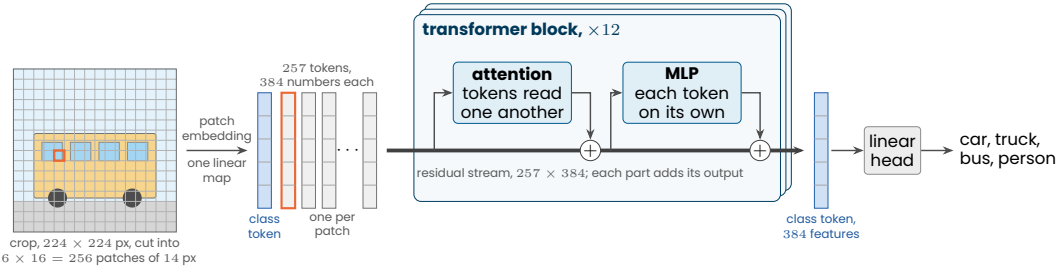
Where the model fails. At 2 bits, with the best step for a Gaussian input, 40% of the inputs fall in the dead zone ($e = -x$) and 6% are clipped. Such errors depend on x, and a minimum-error step shrinks the signal (backup, "A step chosen for minimum error attenuates").

4. PTQ and QAT in a network

Sections 2 and 3 treated one quantizer. A ViT holds dozens, set after training or during it · Notes §5–6

A vision transformer in one picture

Patches become tokens, and twelve blocks refine them

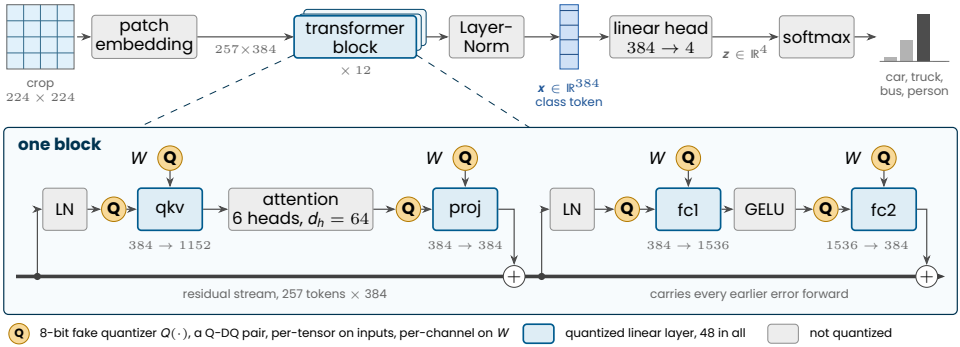


A vision transformer (ViT) turns an image into a sequence of tokens and refines them all together. Attention mixes information across tokens; the MLP, a small two-layer network, transforms each token alone. The head reads only the class token.

ViT-S/14 is the small ViT with 14-pixel patches, the road-scene study's encoder. Most CLIP-family encoders of the VLM study are larger ViTs, such as ViT-B/32 and ViT-L/14. Section 7 introduces CLIP and DINOv2.

Where the quantizers sit in a ViT

96 fake quantizers on the 48 linear layers of DINOv2 ViT-S/14



The road-scene study's classifier is the DINOv2 ViT-S/14 of the last slide. Its 48 linear layers hold 98.2% of the parameters and about 90% of the multiply-accumulates (counted from shapes); their 96 quantizers leave errors in the residual stream. LayerNorm (layer normalization, LN) and GELU (Gaussian error linear unit) are not quantized.

PTQ and QAT share a trunk

Notebook Parts 1 and 5

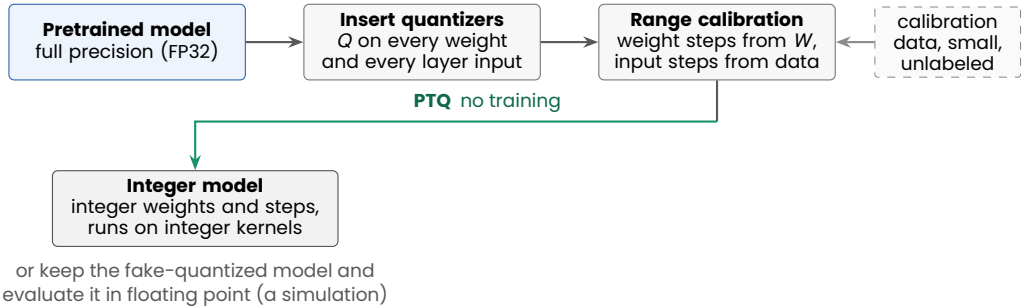
Then post-training quantization exports, and QAT keeps training



PTQ and QAT share a trunk

Notebook Parts 1 and 5

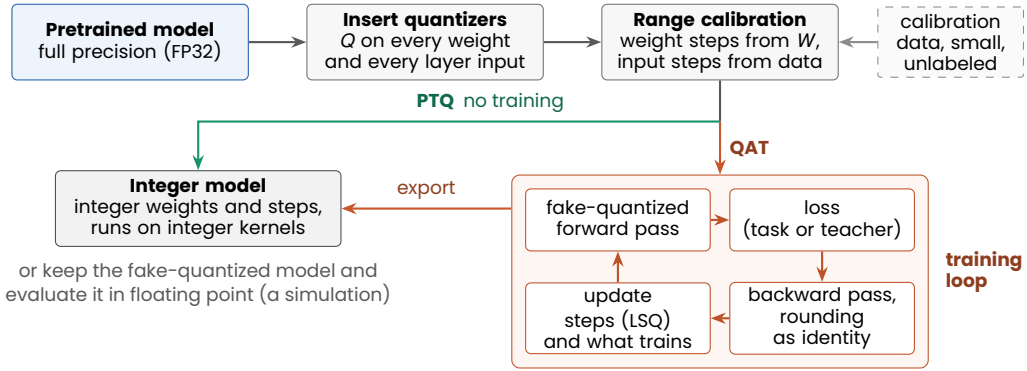
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PTQ and QAT share a trunk

Notebook Parts 1 and 5

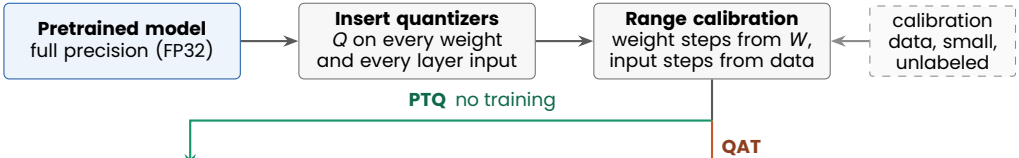
Then post-training quantization exports, and QAT keeps training



PTQ and QAT share a trunk

Notebook Parts 1 and 5

Then post-training quantization exports, and QAT keeps training

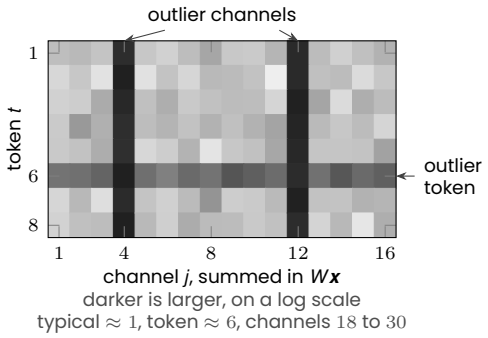


or keep the fake-quantized model and evaluate it in floating point (a simulation)

Two kinds of QAT. **Full QAT** trains the weights and the steps, often by distillation from the FP32 model, so the feature residual mixes rounding with weight drift. **Frozen-backbone QAT** trains only the steps and a head, so the residual is a pure quantization residual, learned clipping included.

Outlier channels

What an outlier is, for a quantizer



Not the OOD sense. Here the input is ordinary, and an outlier channel is large for every input. In OOD detection an outlier is an unusual image or patch (section 8).

What an outlier is. A value far above the typical values that share its step, measured by the peak-to-rms ratio, about 3.2 for 384 Gaussian values.

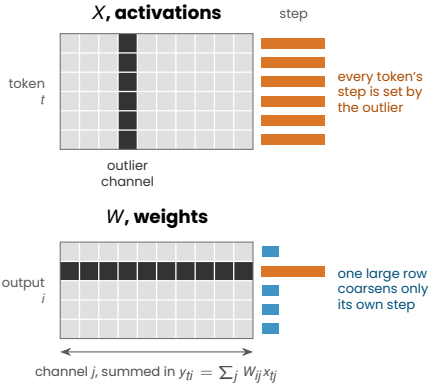
Tokens and channels. A *token* is the vector of one patch (or the class token), a row of the map. A *channel* is one of its 384 coordinates, the same learned feature at every token, not a color.

Three kinds of outlier

- A *rare value* is large in one input only, and can be clipped.
- An *outlier channel* is large at most tokens (Dettmers et al., 2022; Xiao et al., 2023).
- An *outlier token* is large in all its channels (Darcet et al., 2024).

Why outliers hurt activations most

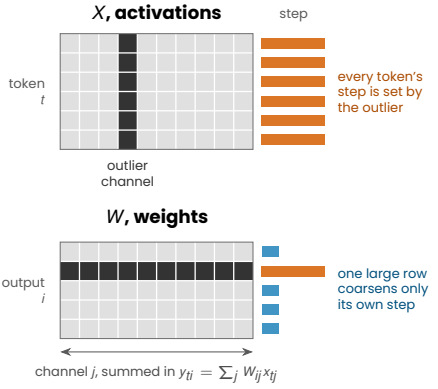
One large channel sets the step of all the others



The granularity rule. A step may depend on any index except j, the one summed in $y_{ti} = \sum_j W_{ij} x_{tj}$. Rows of W and tokens of X may take their own steps; channels may not.

Why outliers hurt activations most

One large channel sets the step of all the others

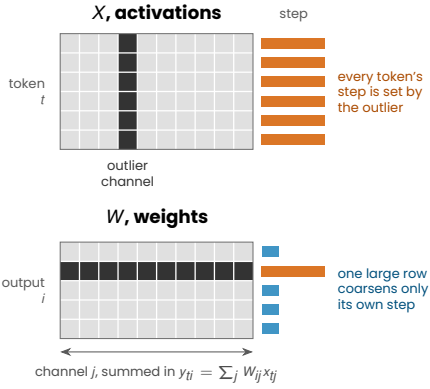


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Why activations. A large weight row coarsens only its own step. An outlier channel is summed, so it sets the step of every token.

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Why activations. A large weight row coarsens only its own step. An outlier channel is summed, so it sets the step of every token.

The cost. With max-abs, an outlier of size M sets the step $M/q_{\max}$, so a typical value of size a spans $q_{\max}/(M/a)$ steps and rounds to zero once $M/a > 2q_{\max}$.

bits	$q_{\max}$	zero once $M/a >$	steps at $M/a = 50$
8	127	254	2.54
6	31	62	0.62
4	7	14	0.14

Post-training methods, by the problem they solve

In red, the eight PTQ methods of the VLM study

Set the range
Simple MinMax (Jacob et al., 2018), one step from the extremes
ACIQ (Banner et al., 2019), a clip value from a Laplace fit

Keep outlier channels in 16 bits
LLM.int8() (Dettmers et al., 2022), a 16-bit product for the outlier dimensions, 8 bits for over 99.9% of the values

Move their scale into the weights
SmoothQuant (Xiao et al., 2023), per-channel smoothing
Outlier Suppression (Wei, Zhang, et al., 2022), moves LayerNorm's γ

The PTQ suite of the VLM study (Bouguerra et al., 2026a, Sec. 3.1); the grouping follows the lecture notes, Table 4.

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Rotate them away
QuaRot (Ashkboos et al., 2024), random orthogonal in the VLM study
SpinQuant (Liu et al., 2025), learned rotations

Choose the rounding of each weight
AdaRound (Nagel et al., 2020), per layer
BRECQ (Li et al., 2021), per block
QDrop (Wei, Gong, et al., 2022)
GPTQ (Frantar et al., 2023)

Protect or rotate weights at ≤ 4 bits
AWQ (Lin et al., 2024), salient channels
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The PTQ suite of the VLM study (Bouguerra et al., 2026a, Sec. 3.1); the grouping follows the lecture notes, Table 4.

Post-training methods, by the problem they solve

In red, the eight PTQ methods of the VLM study

Set the range
Simple MinMax (Jacob et al., 2018), one step from the extremes
ACIQ (Banner et al., 2019), a clip value from a Laplace fit

Keep outlier channels in 16 bits
LLM.int8() (Dettmers et al., 2022), a 16-bit product for the outlier dimensions, 8 bits for over 99.9% of the values

Move their scale into the weights
SmoothQuant (Xiao et al., 2023), per-channel smoothing
Outlier Suppression (Wei, Zhang, et al., 2022), moves LayerNorm's γ

Rotate them away
QuaRot (Ashkboos et al., 2024), random orthogonal in the VLM study
SpinQuant (Liu et al., 2025), learned rotations

Choose the rounding of each weight
AdaRound (Nagel et al., 2020), per layer
BRECQ (Li et al., 2021), per block
QDrop (Wei, Gong, et al., 2022)
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Fit the statistics of a ViT
APQ-ViT (Ding et al., 2022)
IGQ-ViT (Moon et al., 2024)
PTQ4ViT (Yuan et al., 2022), twin uniform grids
RepQ-ViT (Li et al., 2023)

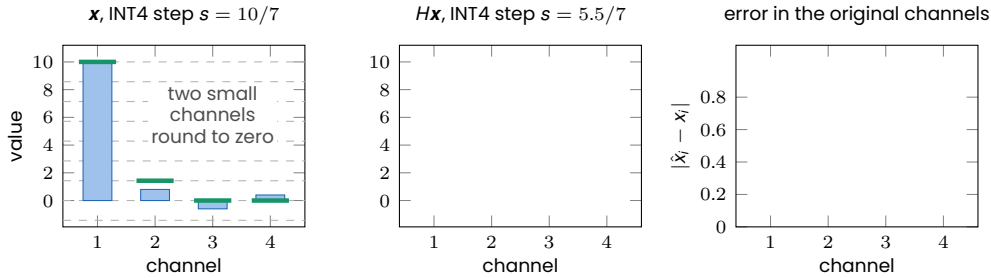
Compensate the error, or search
QwT (Fu et al., 2025), added linear layers
Q-VLM (Wang et al., 2024), block-wise search for VLMs

Stored vectors, such as a KV cache
the keys and values a language model keeps for past tokens
TurboQuant (Zandieh et al., 2026), a rotation, then one precomputed codebook

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Rotation, the idea behind QuaRot and Rotation+LSQ

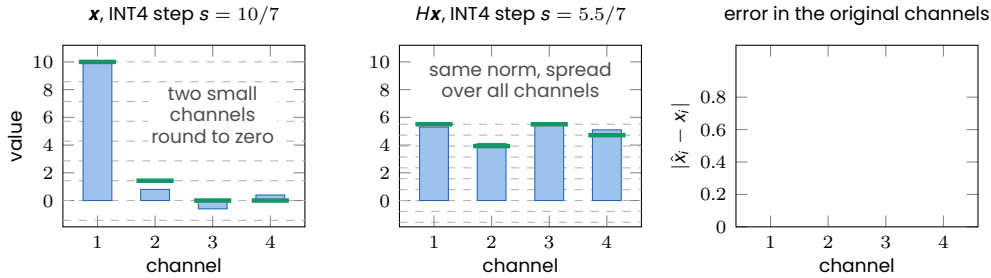
Spread an outlier channel, shrink the step



$W\boldsymbol{x} = (WR^\top)(R\boldsymbol{x})$ for any $R^\top R = I$, so a rotation is free in exact arithmetic. Max-abs sets the step from the peak, and the Hadamard H keeps the norm but lowers the peak-to-rms ratio, from 2 to 1 for $(10, 0, 0, 0)$. QuaRot (Ashkboos et al., 2024) builds its rotations from Hadamard matrices; the VLM study uses random orthogonal ones.

Rotation, the idea behind QuaRot and Rotation+LSQ

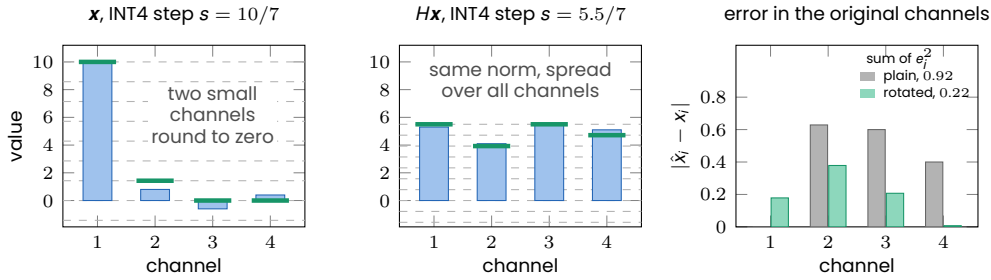
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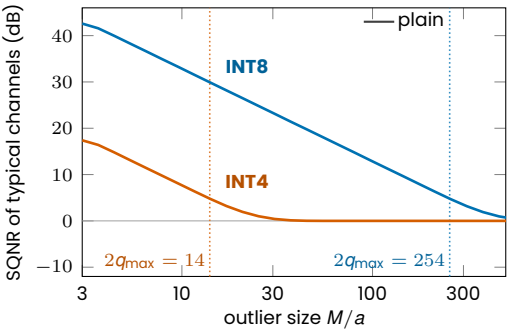
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INT4 here, INT8 in the VLM study's grid

The study keeps activations at 8 bits. The 4-bit part is the weights, and $WR^\top$ rotates each weight row too.

Why a rotation works in general

One outlier channel in a 384-channel token

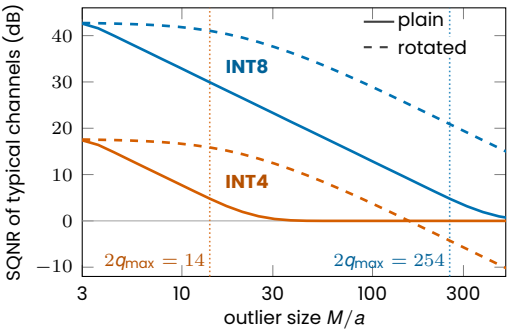


Gaussian coordinates. A random rotation sends $\boldsymbol{x}$ in a uniformly random direction, so every coordinate is close to $\mathcal{N}(0, \|\boldsymbol{x}\|^2/d)$, and the peak-to-rms ratio falls to about 3.2 at $d = 384$, whatever $\boldsymbol{x}$ was.

Synthetic model, max-abs with one step per token; simulation in the lecture notes, §6.2.

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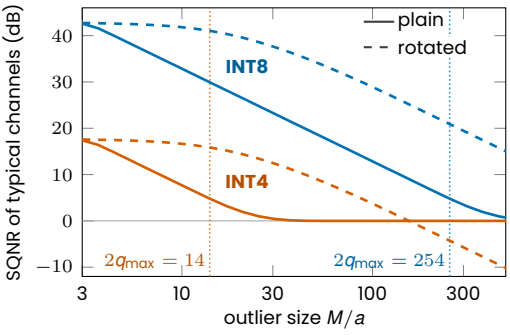
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At $M/a = 50$, INT8 rises from 18.9 to 34.3 dB and INT4 from erased to 9.1 dB. Every row of $WR^\top$ gains in the same way. QuIP (Chee et al., 2023) calls this incoherence, a bound on the peak-to-rms ratio; QuIP# (Tseng et al., 2024) adds a Hadamard transform and an E_8 codebook.

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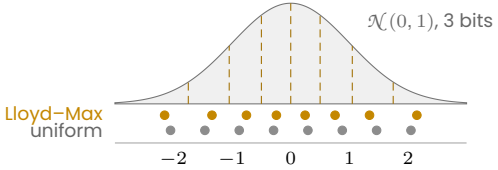
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Limits. The energy stays: an outlier token stays large, and past $M/a \approx 150$ at INT4 the rotated error exceeds plain max-abs. After a GELU, R runs online, cheap with a fast Hadamard transform ($d \log_2 d$ operations).

Synthetic model, max-abs with one step per token; simulation in the lecture notes, §6.2.

After the rotation, a codebook

Lloyd–Max, TurboQuant, and what a codebook costs



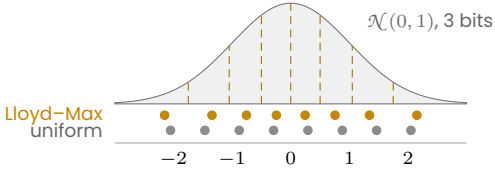
A rotation makes the density known in advance, so the codebook is computed once. Each threshold sits midway between two levels, each level at the mean of its cell (Lloyd, 1982; Max, 1960).

SQNR (dB), Gaussian	2 bits	3 bits	4 bits
Lloyd–Max codebook	9.30	14.62	20.22
best uniform grid	9.25	14.27	19.38
bound $6.02\,b$	12.04	18.06	24.08

SQNR by numerical integration, lecture notes §6.3; the bound from Zandieh et al. (2026, arXiv version, Thm. 3).

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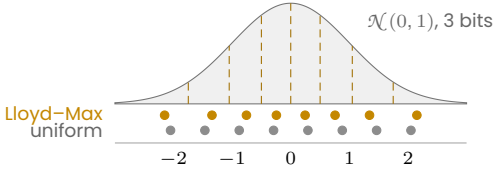
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TurboQuant (Zandieh et al., 2026)

For stored vectors, such as a key–value cache. One random rotation, then every coordinate through one precomputed Lloyd–Max codebook, with no calibration data. The error stays within $\sqrt{3}\pi/2 \approx 2.7$ of a lower bound for any quantizer. A minimum-error codebook shrinks inner products, by $2/\pi$ at 1 bit, so a 1-bit QJL stage on the residual removes that bias. Reported on LongBench: “absolute quality neutrality with 3.5 bits per channel”.

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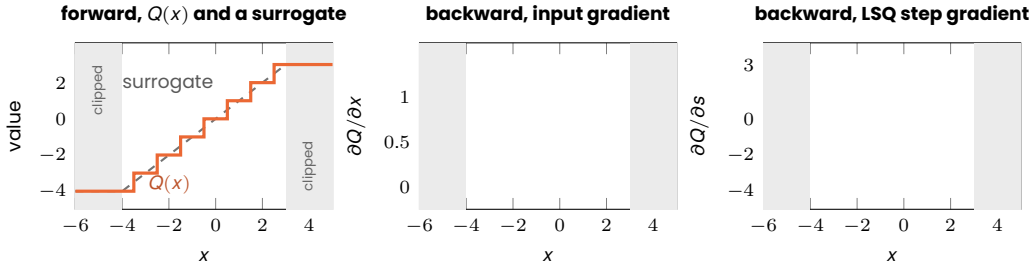
For a vehicle. A codebook index is not a multiple of one step, so the integer kernel of section 1 does not apply. It saves memory and bandwidth, not arithmetic. QuaRot (Ashkboos et al., 2024) performs all its matrix products in 4 bits.

SQNR by numerical integration, lecture notes §6.3; the bound from Zandieh et al. (2026, arXiv version, Thm. 3).

Training through a staircase

Notebook Part 1

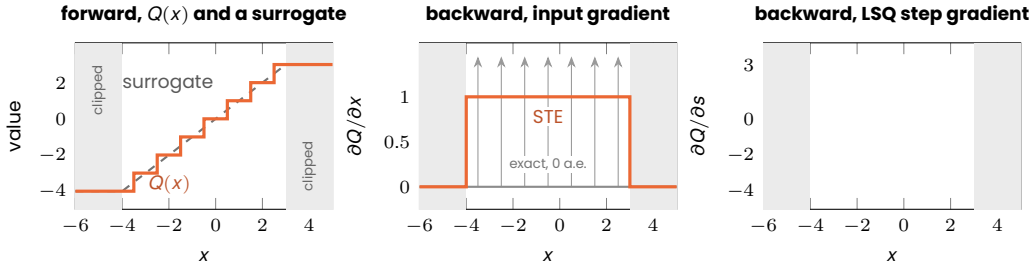
The straight-through estimator, and a learned step size (LSQ)



Training through a staircase

Notebook Part 1

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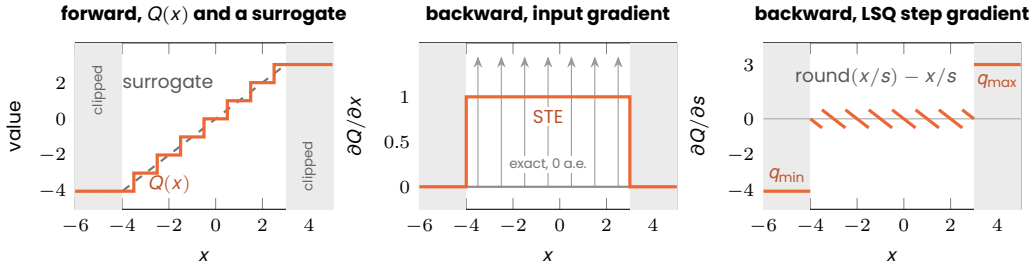


STE (Bengio et al., 2013). Treat round as the identity in the backward pass, so $\partial Q/\partial x \approx 1$ inside $(sq_{\text{min}}, sq_{\text{max}})$ and 0 outside. In code, `(x.round() - x).detach() + x`.

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LSQ (Esser et al., 2020). $\partial Q/\partial s = \text{round}(x/s) - x/s = e/s$ inside, q_{min} or q_{max} outside, times $g = 1/\sqrt{Nq_{\text{max}}}$ for N weights or features. For a squared error, rounding pulls the step down and clipping pushes it up.

Eight quantization-aware variants

What trains, and what each method adds

method	what trains	idea
STE QAT (Courbariaux et al., 2015)	weights	rounding passes gradients as if it were the identity
LSQ (Esser et al., 2020)	weights and steps	the step is a trained parameter
Rotation+LSQ (Bouguerra et al., 2026a)	weights and steps, rotated	the VLM study's own pairing of a QuaRot rotation (Ashkboos et al., 2024) (outliers) with LSQ (step)
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QAT-LoRA (Xu et al., 2024)	low-rank adapters	quantization-aware low-rank adaptation

The first three each run in two distillation regimes, which makes eight. In *distillation* the quantized student learns to match a teacher, presumably the FP32 model. *Contrastive only* uses a contrastive loss, as CLIP does; *hybrid* adds a feature MSE, a distance to the teacher's features.

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Two kinds of QAT

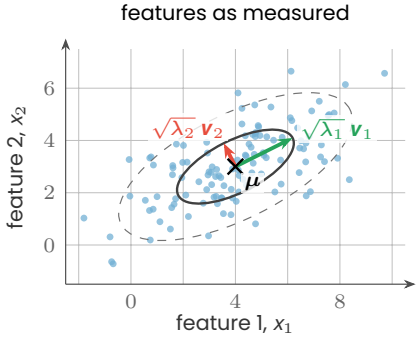
In the VLM study, QAT trains the weights as well as the steps (Bouguerra et al., 2026a, App. C.2). In the road-scene study the backbone is frozen, and only the LSQ steps and the head train (Bouguerra et al., 2026b).

5. Features and their spectrum

The noise of section 3 is the same in every direction. Features are not, so the direction matters · Notes §7–8

A representation and its principal directions

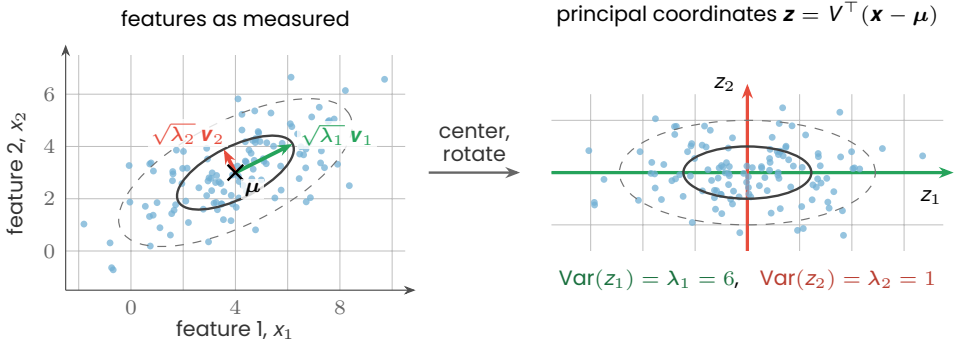
Covariance, eigenvectors, and the SVD that computes them



Stack N feature vectors as the rows of $X \in \mathbb{R}^{N\times d}$ and center them, $X_c = X - \mathbf{1}\boldsymbol{\mu}^\top$. The covariance $C = \frac{1}{N} X_c^\top X_c = V \Lambda V^\top$ has the principal directions $\boldsymbol{v}_k$ as the columns of V, and the variances λ_k along them, 6 and 1 here.

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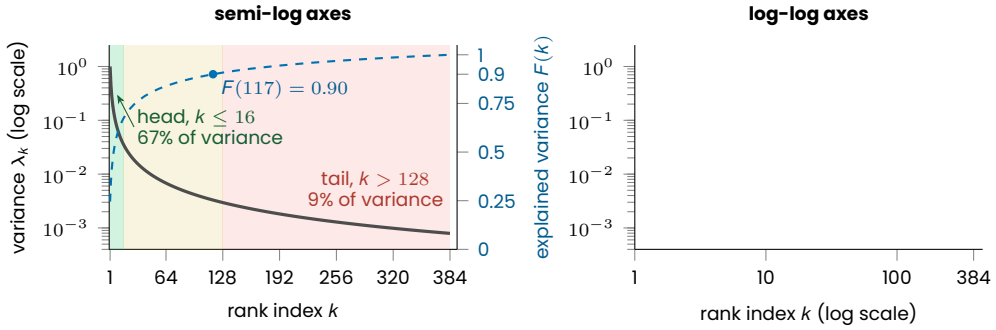
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The SVD computes the same thing. The singular value decomposition of the centered matrix, $X_c = U \Sigma V^\top$, gives the same directions V, with $\lambda_k = \sigma_k^2 / N$. Software takes this route rather than forming C.

The spectral head and tail

Notebook Part 3

Principal directions of the encoder's features, sorted by variance

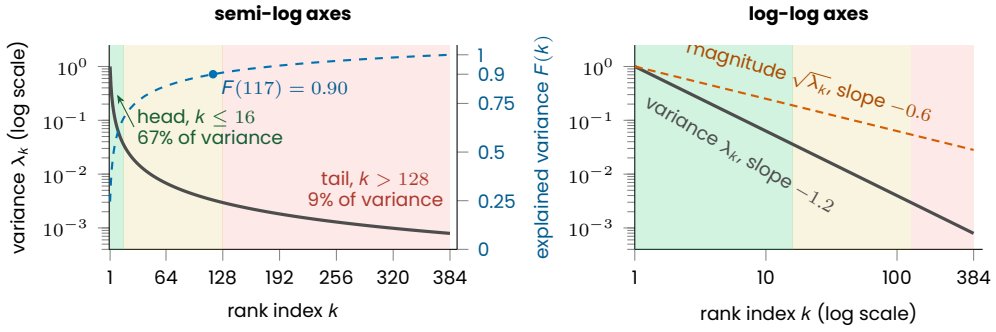


Rank index k**.** $k = 1$ is the largest variance. Small k is the spectral head, large k the tail, and a "high-rank" direction is a *low*-variance one. Plots that count from 0 show $j = k - 1$.

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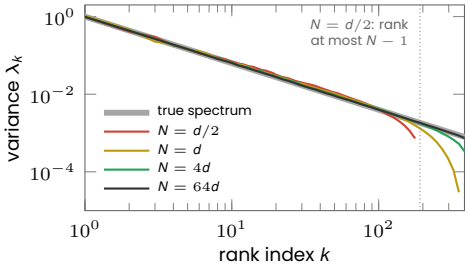
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How many samples does a spectrum need?

The ratio N/d matters, not the size of the dataset

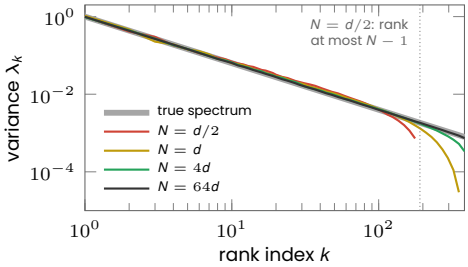


Synthetic: true $\lambda_k = k^{-1.2}$, $d = 384$, Gaussian features, one draw for each N.

- **Below $N = d$ the tail is missing.** N centered vectors span at most $N - 1$ directions.
- **The head needs few samples, the tail many.** At $N = 4d$ the last eigenvalues come out about half their size, at $N = 64d$ within 6%.

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- **The head needs few samples, the tail many.** At $N = 4d$ the last eigenvalues come out about half their size, at $N = 64d$ within 6%.
- **VLM study.** The ImageNet-1k validation set, $50\,000$ images, against $d = 768$ for a ViT-B, so $N/d \approx 65$ (App. D) (Bouguerra et al., 2026a).
- **Road-scene study.** 1462 clean test crops against $d = 384$, so $N/d \approx 3.8$ (Sec. III, VII) (Bouguerra et al., 2026b).
- The whole dataset is cheap, since the $d \times d$ covariance adds up in one pass.

Small variance is not small importance

And a fixed basis is not a recomputed one

The bus crop in four principal coordinates

	z_1	z_2	z_3	z_4	cosine
crop	1.8	0.9	0.3	0.1	
bus prototype	2.0	1.0	0.4	0.3	0.996
truck prototype	2.0	1.0	−0.4	−0.3	0.932

A prototype is the mean feature of a class. The two classes agree on the spectral head and differ only in the tail. Keep $k \leq 2$ and both cosines are 1.000, a tie. Dropping 2.4% of the squared length removed all the evidence.

Synthetic numbers, from the lecture notes.

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Fixed basis. Measure new features X', quantized or blurred, on the FP32 directions, $\boldsymbol{v}_k^\top C' \boldsymbol{v}_k$. Rank index k keeps meaning "the FP32 direction $\boldsymbol{v}_k$".

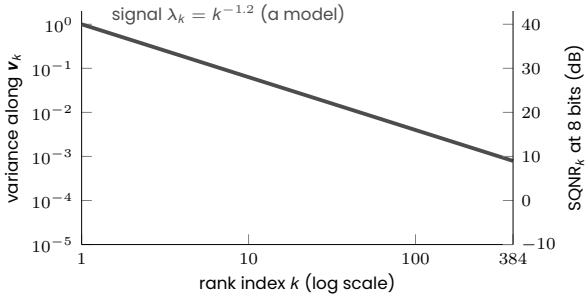
Recomputed basis. Diagonalize C' itself. Its top k directions hold the most variance that any k directions can (Ky Fan's principle), so a recomputed spectrum always looks at least as concentrated in its head, when both are centered at the new mean.

Synthetic numbers, from the lecture notes.

Noise against signal, per rank index

Notebook Part 3

The organizing picture, and the measurement both papers use



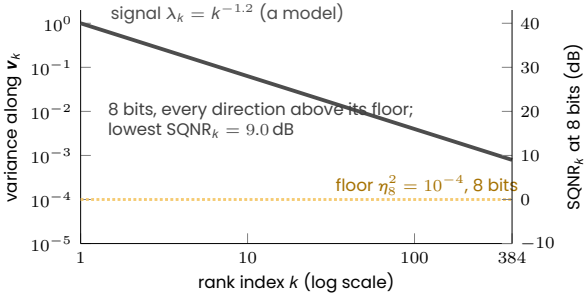
The flat floor needs one step for all coordinates, the high-resolution regime, and uncorrelated errors.

A model. Error of equal power η^2 in every direction lifts every λ_k to $\lambda_k + \eta^2$, a flat floor, about 6 dB lower per added bit.

Noise against signal, per rank index

Notebook Part 3

The organizing picture, and the measurement both papers use



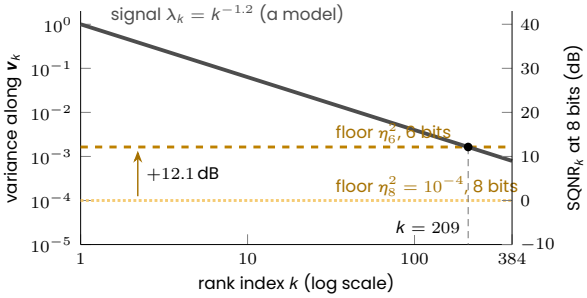
The flat floor needs one step for all coordinates, the high-resolution regime, and uncorrelated errors.

A model. Error of equal power η^2 in every direction lifts every λ_k to $\lambda_k + \eta^2$, a flat floor, about 6 dB lower per added bit.

Noise against signal, per rank index

Notebook Part 3

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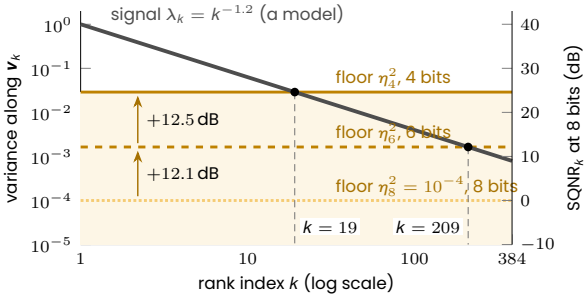
The flat floor needs one step for all coordinates, the high-resolution regime, and uncorrelated errors. A dot marks 0 dB, where half the variance along $\boldsymbol{v}_k$ is still signal.

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The measurement (notes, Def. 8.1)

For a residual E, quantized or blurred minus FP32, on the *fixed* FP32 basis

$$e_k = \tfrac{1}{N}\|E\boldsymbol{v}_k\|^2, \qquad \rho_k = \frac{e_k}{\lambda_k},$$
$$\mathrm{SQNR}_k = -10\log_{10}\rho_k \text{ dB}.$$

So e_k is the residual energy along $\boldsymbol{v}_k$, and ρ_k that energy over λ_k.

Real residuals are not flat. The picture organizes the evidence; it does not explain a gain.

6. Images, blur, and frequency

Quantization noise is one perturbation. Blur, the other, has its own spectrum, over pixels · Notes §9

What degrades a camera image

Synthetic corruption suites, and real adverse weather

benchmark	task	what it applies
ImageNet-C (Hendrycks & Dietterich, 2019)	classification	15 corruptions (noise, blur, weather, digital) at 5 severities
Pascal-C, COCO-C, Cityscapes-C (Michaelis et al., 2019)	2D detection	the same corruptions, on a driving dataset among others
Kamann and Rother (2020)	segmentation	corrupted Cityscapes, PASCAL VOC, ADE20K
KITTI-C, nuScenes-C, Waymo-C (Dong et al., 2023)	3D detection	27 corruptions of camera and LiDAR inputs
DAWN (Kenk & Hassaballah, 2020), ACDC (Sakaridis et al., 2021)	detection, segmentation	real fog, rain, snow, sandstorm, and night

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- Detectors keep only 30 to 60% of their clean performance on corrupted images (Michaelis et al., 2019), and camera-only 3D detectors are the most vulnerable (Dong et al., 2023).
- The usual remedy, training with augmentation (Hendrycks et al., 2020), has to anticipate the degradations the vehicle will meet.

Why blur, and not the other degradations?

Its sources, what it removes, and why study one in depth

- **Out of focus.** A point spreads over a disk.
- **Motion.** A point spreads along a segment that grows with the exposure, so night makes it worse.
- **Water and dirt** on the lens add halos and smears (Ceccarelli & Secci, 2023).

A car crossing at 50 km/h, 20 m away (synthetic)

With a focal length of 1000 px, the image moves at $1000 \cdot 13.9/20 \approx 694$ px/s, so a 20 ms night exposure smears it over about **14** px.

All of these remove fine detail first, the detail of far objects, sign text, and truck against bus. A 10-px box blur left no tested detector above 0.6 on a 0-to-1 detection score, though the vehicles stayed visible (Ceccarelli & Secci, 2023).

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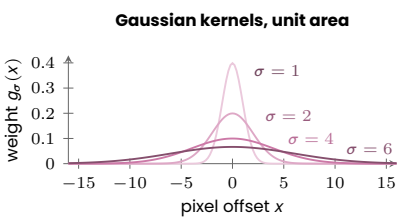
- **Many failures end in blur**, the road-scene study's reason for its choice (Bouguerra et al., 2026b). The catalog it cites lists 26 failures, unranked.
- **One knob.** A Gaussian blur has one parameter, σ, is deterministic, and can be added on top of real bad weather.
- **It fits the spectral account.** A literal low-pass filter, so it tests the spectral story of sections 5 and 6 directly.
- **The others are not ignored.** Real weather is the baseline, noise and contrast are controls, and the VLM study's suite adds noise, brightness, and contrast. Glare, night, and occlusion stay untested.

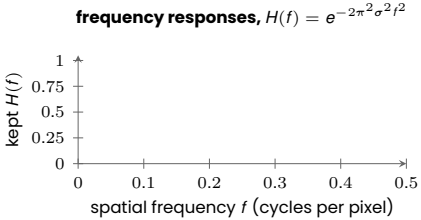
Blur is a low-pass filter

Notebook Part 4

Only σ over the period of a detail matters

Blur averages neighboring pixels. Seen as a sum of stripes of frequency f, it multiplies each stripe by its response $H(f)$, so the fine stripes vanish first.



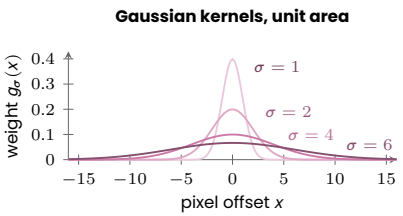


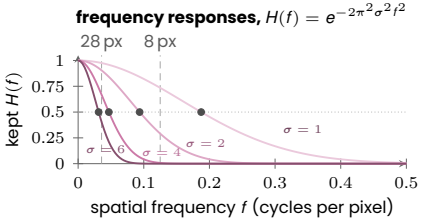
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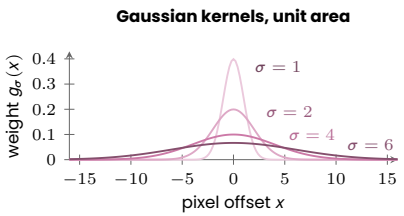


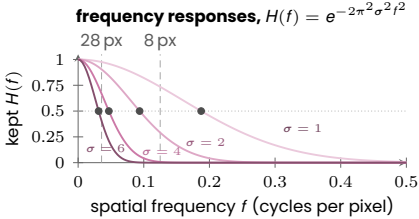
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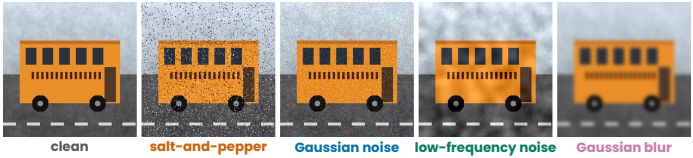


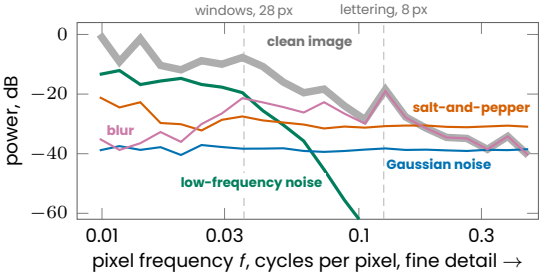




Low- and high-frequency corruptions

Where each one acts in frequency, on a synthetic image





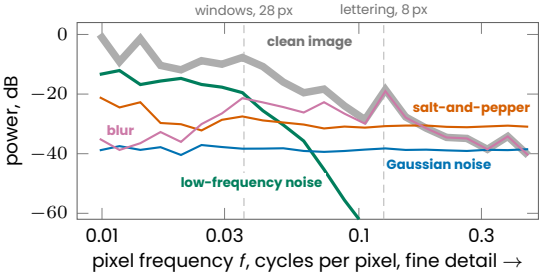
Colored lines show the change, corrupted minus clean.

- **High-frequency noise** is flat, so it rises above the weak fine detail and buries it.
- **Low-frequency noise**, as strong as the Gaussian noise, sits at coarse scales, where the image has its power. It changes shading, not detail.
- **Blur** removes fine detail and adds nothing. Its change never exceeds the image.

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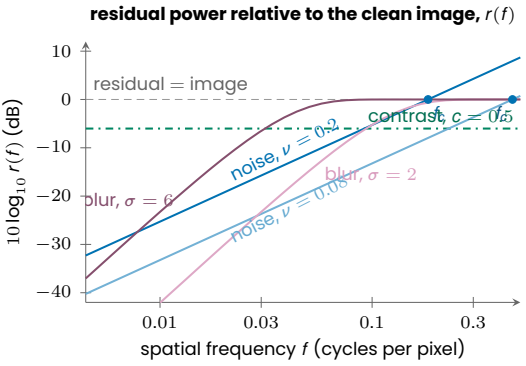


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- **Blur** removes fine detail and adds nothing. Its change never exceeds the image.
- Gaussian-noise training helps against high-frequency corruptions and hurts against low-frequency ones (Yin et al., 2019).

Relative to the image, as a model

The ratio $r(f)$, and why a blur σ needs its image

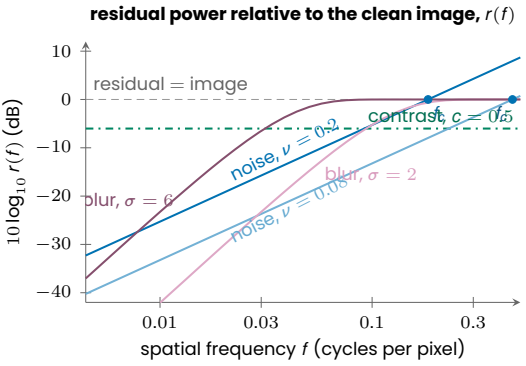


Model image $P(f) = A/f^2$. The ratio $r(f)$ of residual to clean power is the pixel counterpart of ρ_k.

- **Gaussian blur erases** fine detail, so $r(f) \leq 1$.
- **White noise buries** it, relative to the image, above $f_c = \sqrt{A}/\nu$. Its own spectrum is flat.
- **Contrast** scales every frequency by the same factor.

Relative to the image, as a model

The ratio $r(f)$, and why a blur σ needs its image



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- **Contrast** scales every frequency by the same factor.
- **σ is in pixels of the image it acts on.** Enlarging by m turns σ into σ/m, so blurring a 224-pixel crop and a 640-pixel frame are different corruptions.

Two spectra, and what “high” means in each

Frequencies of one image, variances of many features

	image spectrum	feature spectrum
describes	one image	many feature vectors
axis	spatial frequency f, cycles per pixel	rank index k, from the largest variance
basis	fixed sinusoids, one for all images	learned principal directions $\boldsymbol{v}_k$
“high” means	fine detail	a low-variance direction, the tail
filter	blur, a literal low-pass filter $H(f)$	none, nothing multiplies the rank axis

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basis	fixed sinusoids, one for all images	learned principal directions $\boldsymbol{v}_k$
“high” means	fine detail	a low-variance direction, the tail
filter	blur, a literal low-pass filter $H(f)$	none, nothing multiplies the rank axis

Three things called “spectral”

the *image spectrum* · the *feature spectrum* · *spectral bias*, the tendency of networks to fit low-frequency functions of their input first (Rahaman et al., 2019).

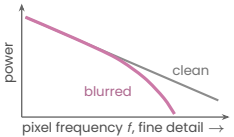
The VLM study’s “spectral filtering” means a low SQNR_k in the feature tail, not a filter. Nothing removes the tail; in the paper’s Fig. 8b its magnitude even rises for PTQ and QAT (Bouguerra et al., 2026a).

How the two spectra connect

Pixel frequency is not feature rank

Notebook Part 4

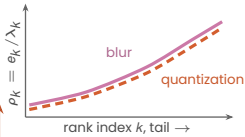
1. Blur, a literal filter on f



2. encoder ϕ, nonlinear

quantizers inside ϕ

3. Residuals on clean v_k



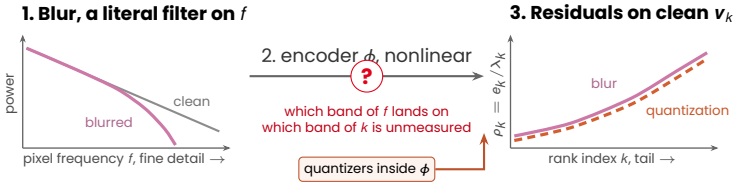
4. The road-scene study compares the two profiles rank by rank.

Curves drawn for shape, not read off a paper.

How the two spectra connect

Pixel frequency is not feature rank

Notebook Part 4



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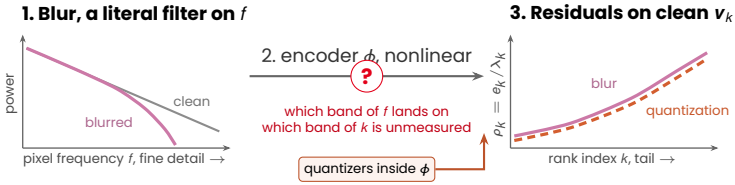
Curves drawn for shape, not read off a paper.

Tempting, not measured. Fine detail would sit in the tail. Networks fit coarse structure first (Rahaman et al., 2019) and can use fine cues people barely see (Wang et al., 2020), yet neither says where a band of f lands on k. Noise and contrast act at other f than blur, and their residuals line up with the quantization residual about as well (Bouguerra et al., 2026b).

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Pixel frequency is not feature rank

Notebook Part 4



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A direct test

Add noise of fixed energy, one octave of f at a time, and compute the per-rank energy e_k on the clean basis. If the pairing holds, the profile slides toward the tail as the octave nears $f = 0.5$, the finest detail.

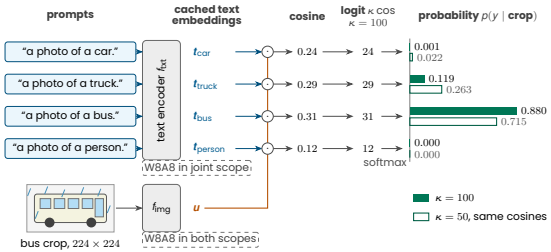
7. Perception models

Sections 4 to 6 used two models. Here is what each one is, and how it classifies · Notes §10

Vision-language models in the vehicle

CLIP classifies from text prompts

CLIP maps images and text into one space. *Zero-shot*, it picks the class whose prompt lies closest to the image, with no training on the task.



In the vehicle. VLMs are entering driving stacks, for perception, understanding, and planning (Zhou et al., 2024).

Two scopes. *Visual-only* keeps the text encoder in FP32, at no cost for cached prompts. *Joint* quantizes both encoders.

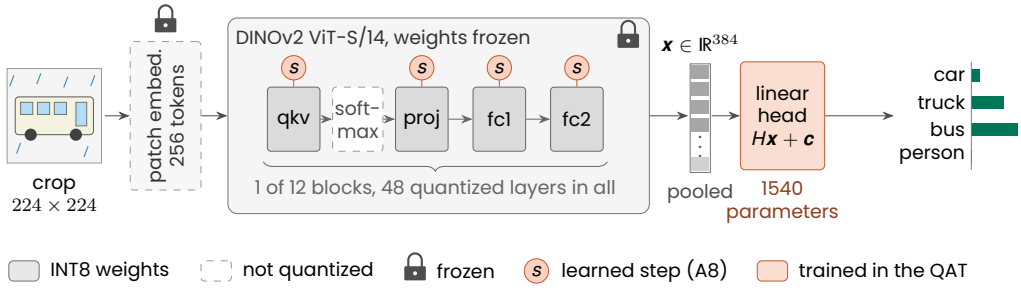
The logit scale κ, a learned temperature, changes the confidence, never the winner. Refitting it repairs confidence, never accuracy.

DINOv2 with a linear head

Notebook Part 5

And what its quantization-aware training trains

DINOv2 is an image encoder trained without labels; a linear head reads its features.



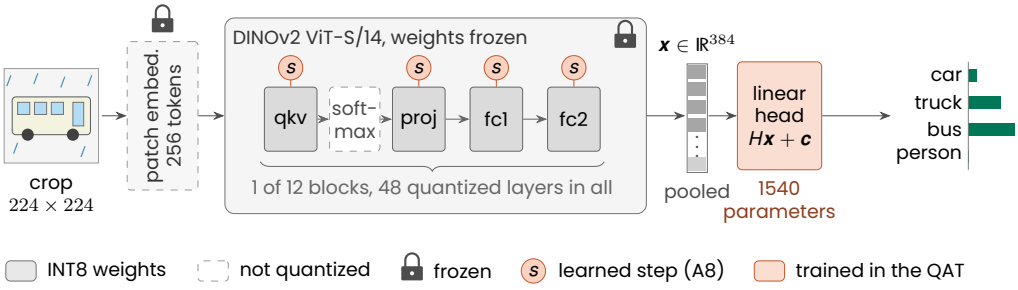
Frozen-backbone QAT trains the head,
$4 \times 384 + 4 = 1540$ numbers, and 48 activation steps, so 1588 numbers, 7.3×10^{-5} of the 21.6M parameters of ViT-S/14.

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The FP32 reference gets the same extra training iterations, or the comparison would mix quantization with extra training.

8. Reliability and fair comparisons

Beyond top-1 accuracy, and how to compare two models · Notes §12–13

Reliability beyond top-1

Five questions, their metrics, and what they mean on the road

dimension	the question	metric	on the road
corruption robustness	does performance hold as the image degrades?	accuracy per severity	blur, fog, a dirty lens
natural covariate shift	does it hold on new cities, cameras, seasons?	performance on a shifted set	a new market or camera
probability calibration	is a confidence of 0.8 right 80% of the time?	expected calibration error (ECE)	confidences fed to fusion and planning (Feng et al., 2022)
the unknown (OOD)	does it flag what it was never taught?	AUROC, FPR@95	debris or an unusual vehicle (Blum et al., 2021)
spurious reliance	does it read the context instead of the object?	relative spurious gap (RSG)	a bus where buses are rare

OOD, out-of-distribution; AUROC, area under the receiver operating characteristic (ROC) curve; FPR@95, false positive rate at 95% true positive rate.

An AUROC of 0.86 can coexist with an FPR@95 of 0.56, so most unknowns still pass (synthetic model).

Two calibrations. *Range* calibration sets the quantizer steps (section 4); *probability* calibration checks the confidences.

Plain accuracy and balanced accuracy

Two averages of the same per-class recalls

		FP32		W8A8	
class	N_i	correct	recall	correct	recall
car	700	665	95.0%	637	91.0%
truck	150	96	64.0%	102	68.0%
bus	100	48	48.0%	60	60.0%
person	50	35	70.0%	38	76.0%

1000 synthetic crops of blurred road objects, from the lecture notes.

$$\text{Acc} = \sum_i \frac{N_i}{N} r_i, \qquad \text{BAcc} = \frac{1}{K} \sum_i r_i$$

Recall r_i is the fraction of class i classified correctly. Chance level of BAcc is $1/K = 25\%$ for four classes, whatever the class mix.

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Acc			84.4%		83.7%
BAcc			69.25%		**73.75%**

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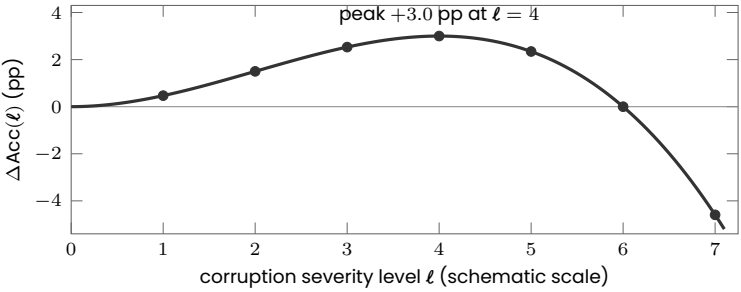
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Plain accuracy falls by 0.7 points while balanced accuracy **rises** by 4.5. One crop of class i is worth $100/(KN_i)$ points, 0.036 for a car and 0.5 for a person.

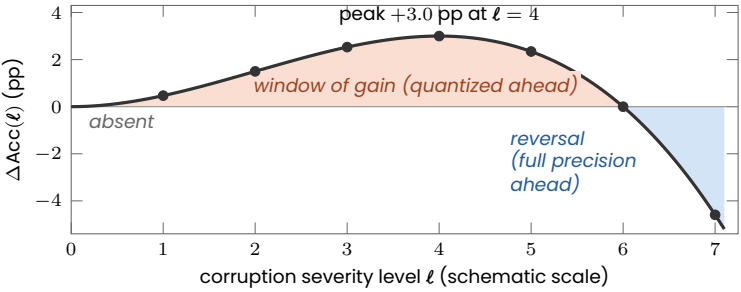
Reading a robustness result

Per severity, and against a matched control



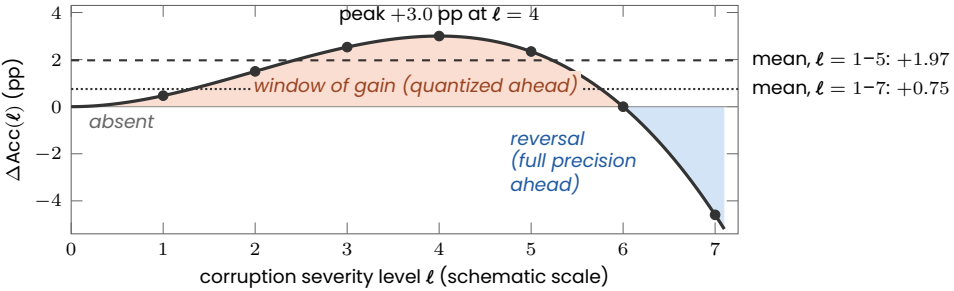
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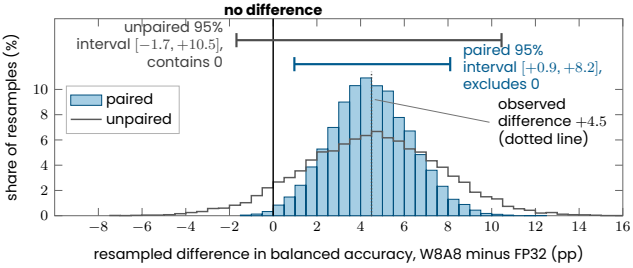
Per severity, and against a matched control



- Report the effect per severity, with any average next to the curve, never instead of it.
- Compare against a control that differs only in quantization: same start, budget, data order, and seeds.

What an error bar covers

Seeds, standard errors, and the paired bootstrap

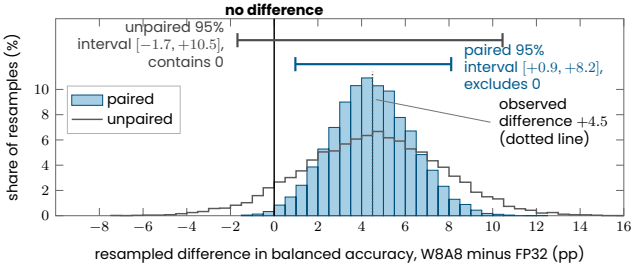


The bootstrap redraws the test crops with replacement, within each class, and rescores, $10\,000$ times. The classifiers are the two synthetic ones of the balanced-accuracy slide.

- **Three seeds** printed as 72.0 ± 1.0 (population SD) give a 95% interval $[69.0,\ 75.0]$ for the mean, three times the printed spread. With two degrees of freedom, $t = 4.3$ multiplies the standard error, 0.71.

What an error bar covers

Seeds, standard errors, and the paired bootstrap

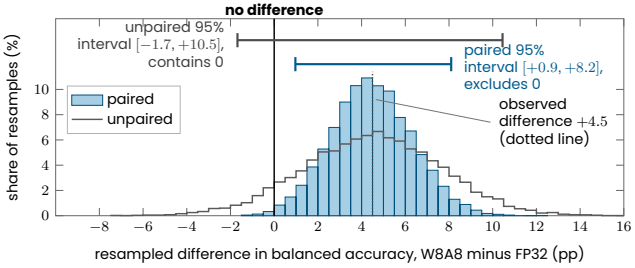


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What an error bar covers

Seeds, standard errors, and the paired bootstrap



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- **Pairing.** Only crops that one model gets right and the other wrong carry the difference, so the paired interval is narrower.
- A crop bootstrap covers new test crops, not new training seeds.

9. The question

Two literatures that grew apart, the gap, and two featured papers · Notes §14

The question of this tutorial

Part II starts here

Can less precise be more reliable?

Can 8-bit quantization, chosen for efficiency, also make a perception model more robust to blur, noise, and bad weather, instead of less?

Two demands that pull apart

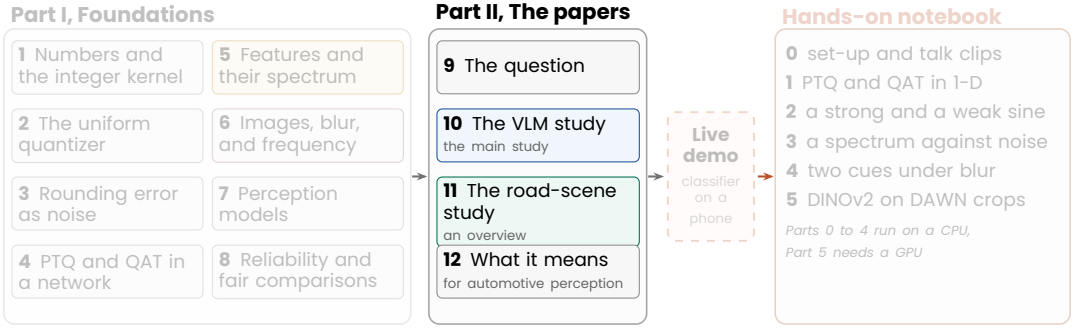
Reliable when the image degrades, and efficient on embedded hardware with tight limits on cost, power, and heat. Quantization serves the second and is widely assumed to hurt the first.

How Part II answers it

The VLM study tests it at scale, the road-scene study on one road-object classifier, and the last section says what it means for a perception team.

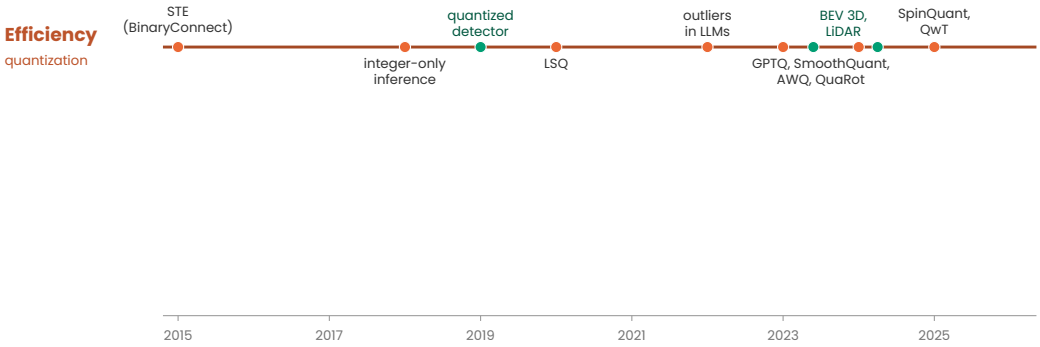
Roadmap, Part II

The tools of Part I, applied to the two studies



Two literatures that grew apart

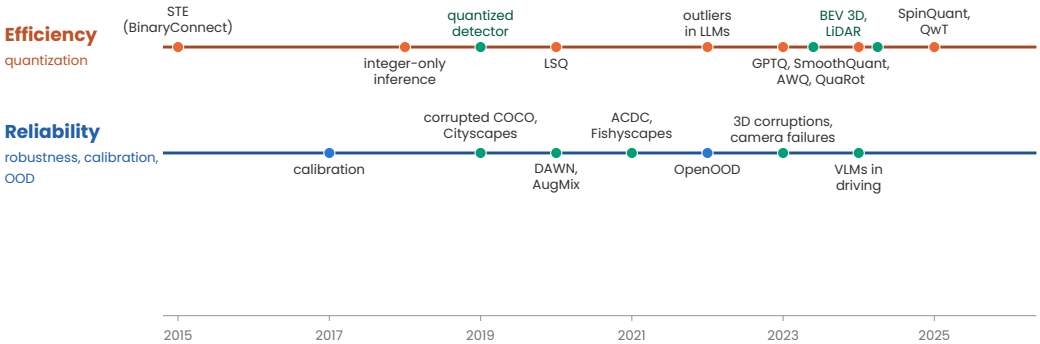
And the thin lane where they met



"These two domains (reliability and efficiency) have evolved in parallel" (Bouguerra et al., 2026a). "The two literatures have run in parallel" (Bouguerra et al., 2026b). Driving-specific works in green.

Two literatures that grew apart

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The gap, and why a gain would be surprising

Noise cannot create information

The gap

No systematic study of how quantization changes the reliability of vision-language models (Bouguerra et al., 2026a).

For driving, no study that matches training budgets or deploys an integer stack (Bouguerra et al., 2026b).

Proposition (notes, 14.1)

Let $\boldsymbol{z}$ be the features, y the label, and $\boldsymbol{n}$ a noise. If $\boldsymbol{n}$ is independent of $(\boldsymbol{z}, y)$, every rule on $\boldsymbol{z} + \boldsymbol{n}$ is matched by a rule on $\boldsymbol{z}$ that draws its own $\boldsymbol{n}'$. The best accuracy from $\boldsymbol{z} + \boldsymbol{n}$ never exceeds the best from $\boldsymbol{z}$.

The gap, and why a gain would be surprising

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So a gain means a premise fails, and in both papers three do.

1. **Garbling.** Rounding acts on every layer's activations, with an error tied to the signal. The quantized features are another function of the image, not a noisy copy.
2. **Optimality.** Neither model is the best rule for blurred images, which appear only at evaluation.
3. **Sameness.** The quantized model is trained around its own perturbation.

Which failure produces the gains is not settled.

Two featured studies, a main one and a road case

Breadth first, then one road-scene classifier

The main study, quantized VLMs at scale

Bouguerra et al. (2026a), ICML 2026, arXiv:2509.21173.

About $708{,}480$ evaluation runs, 10 CLIP-family vision-language models, 16 quantization methods, five reliability dimensions, and a spectral account.

Buys a distribution of outcomes. **Lacks** a matched control.

A road case, with its own talk

Bouguerra et al. (2026b), IEEE ICVES 2026, day 2.

A DINOv2 classifier of DAWN road objects, each variant matched to its FP32 control. Here, an overview; a live demo closes the talk.

Buys attribution. **Lacks** generality.

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The shared claim, in the papers' own words

Quantization "can simultaneously improve accuracy, calibration, OOD detection, and robustness to noise, though not to covariate shift or spurious correlations" (VLM study, Abstract). Classification-stage quantization is "a possible source of corruption robustness rather than a cost" (road-scene study, Sec. IX).

10. Quantized vision-language models at scale

The VLM study, axis by axis, then its spectral account · Notes §15–17

Design, breadth by construction

The grid of App. C.5, and five reliability dimensions

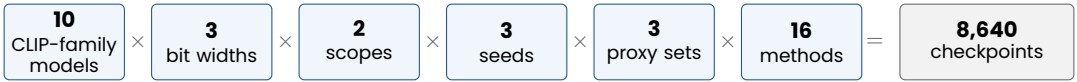
10 CLIP-family models × **3** bit widths × **2** scopes × **3** seeds × **3** proxy sets × **16** methods

A proxy set is 1000 image-caption pairs, from CC3M, YFCC, or SBU, that set the PTQ ranges and train the QAT variants (Sec. 3.1).

Five dimensions (Sec. 1). Accuracy under quantization, calibration (ECE), OOD detection (AUROC, FPR@95), natural covariate shift, and spurious correlations (ΔRSG on CounterAnimal, animal photos in common and in rare backgrounds).

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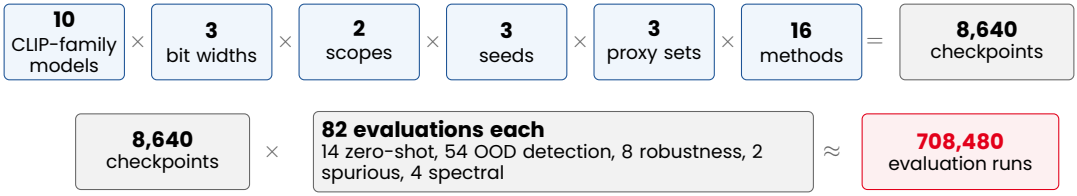


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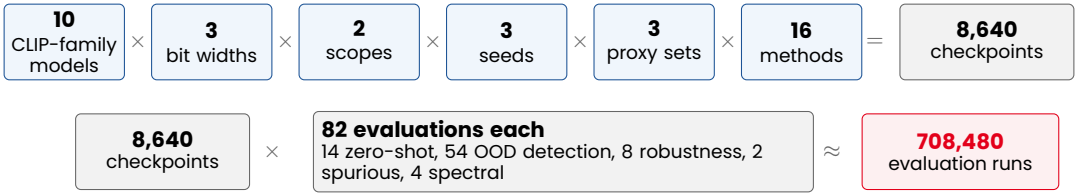


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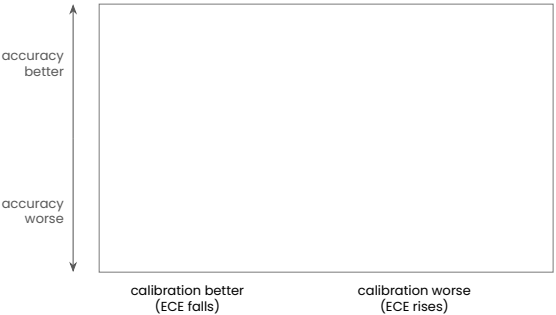
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Read with care. All 10 models are pretrained once, so a claim about models rests on 10 points. All 16 methods are simulated, plus one INT8 engine built with TensorRT, NVIDIA's inference compiler. No FP32 model got the same proxy training (Sec. 3.2).

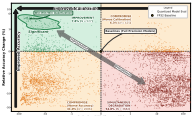
Not systematically destructive

Fig. 1, 6720 “significant” W8A8 runs, read as a two-by-two table



The printed shares and counts of Fig. 1, drawn as a mosaic, areas to scale.

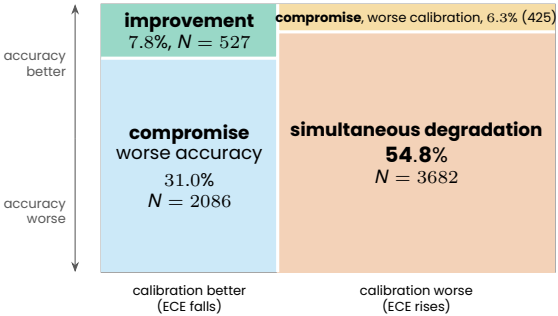
Bouguerra et al. (2026a, Fig. 1).



The paper’s Fig. 1, both axes logarithmic.

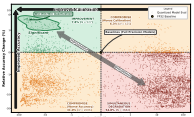
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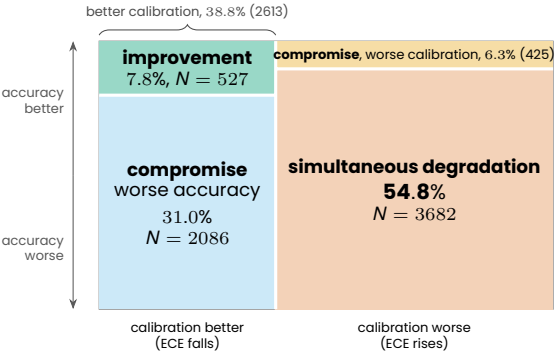


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- **Not systematically destructive.** One run in thirteen improves both; more than a third improve calibration. *(measured)*

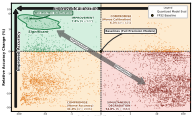
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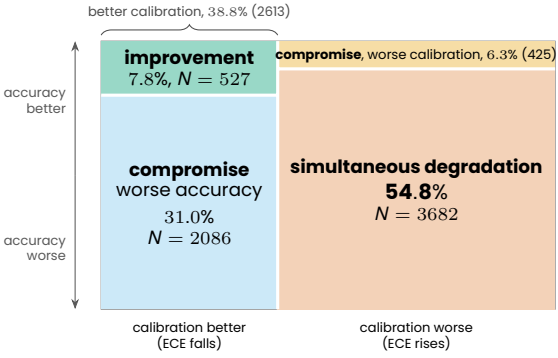


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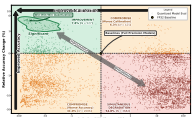
- **Not systematically destructive.** One run in thirteen improves both; more than a third improve calibration. *(measured)*
- **Yet** 54.8% degrade both, and accuracy falls in 85.8% of runs. *(measured)*

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- **Not systematically destructive.** One run in thirteen improves both; more than a third improve calibration. *(measured)*
- **Yet** 54.8% degrade both, and accuracy falls in 85.8% of runs. *(measured)*
- Calibration gains bring accuracy gains twice as often as calibration losses do (20.2% against 10.3%, the notes’ arithmetic).
- The test behind “significant” is not named; every share describes a filtered set.

Bouguerra et al. (2026a, Fig. 1).

Accuracy, calibration, OOD detection

For every number, name the population

Accuracy · Sec. 4.1, Figs. 2, 16

The mean relative accuracy change is negative for all ten models, under PTQ and QAT, and QAT loses less in every model.

PTQ loss tracks loss under weight dropout, $r = 0.91$ over 10 model-level points; the "redundancy" reading is plausible, not isolated.

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Calibration · Sec. 4.2, Figs. 4, 5

Over successful W8A8 runs, those losing under 2% on clean data, relative ECE change $+9\%$ visual-only ($N = 5543$). Joint, $+98\%$ ($N = 3893$, $p = 0.1$), not significant.

Rotation+LSQ -3% and -5% (Fig. 4B).

LAION ECE goes $6.9 \to 3.1 \to 1.1\%$ from FP32 to QAT to a refitted logit scale (Fig. 5). No FP32 model got the same refit, so that comparison is untested.

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OOD detection · Sec. 4.3, Fig. 20A

Relative in-distribution (ID) accuracy and AUROC changes correlate, $r = 0.994$, slope 0.428, over 96 aggregated points, not $466{,}560$ runs. OOD moves less than half as much as accuracy.

Calibration · Sec. 4.2, Figs. 4, 5

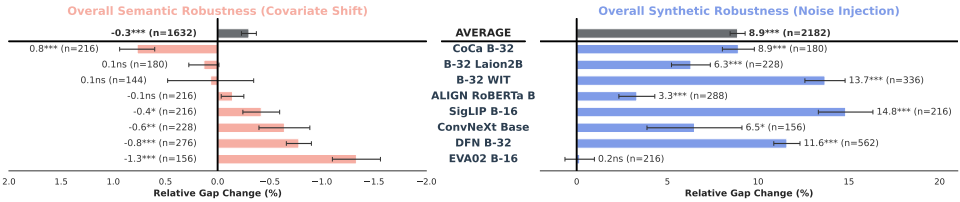
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Corruptions versus natural covariate shift

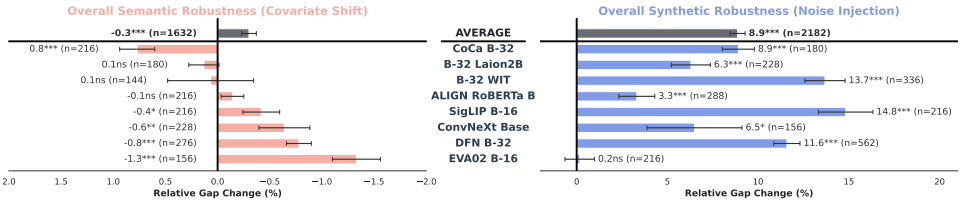
Fig. 7, over successful W8A8 runs



Runs that lose less than 2% on clean data. Natural covariate shift (left) moves by -0.3, $n = 1632$; synthetic corruption (right) by $+8.9$, $n = 2182$, significant for seven of eight models. The left axis runs from $+2$ to -2, so its bars to the right are losses; on the right, bars to the right are gains.

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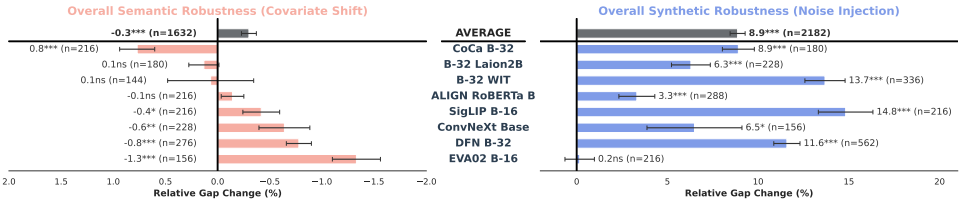
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Three limits

- only runs that kept their clean accuracy,
- a “relative gap change” with no formula, not convertible to points,
- four corruptions pooled, never reported one by one.

Corruptions versus natural covariate shift

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Three limits

- only runs that kept their clean accuracy,
- a "relative gap change" with no formula, not convertible to points,
- four corruptions pooled, never reported one by one.

And the bus in the rain?

The suite holds no rain, snow, or fog (App. C.4). Whether $+8.9$ or -0.3 applies to rain is unknown.

Where quantization hurts

Spurious correlations (Table 1), and 4 bits

△RSG on CounterAnimal, positive is more spurious reliance

method	W8A8	W6A8	W4A8
Simple PTQ	2.6	2.8	12.5
QAT (Contr.)	1.6	2.1	9.1
Rot+LSQ	−0.1	0.2	4.0

Means over 10 runs. Quote the table, not Sec. 4.5's prose, which disagrees with it.

- Every PTQ and QAT cell is positive, several times larger at 4 bits. Rotation+LSQ stays near zero at 8 and 6 bits.

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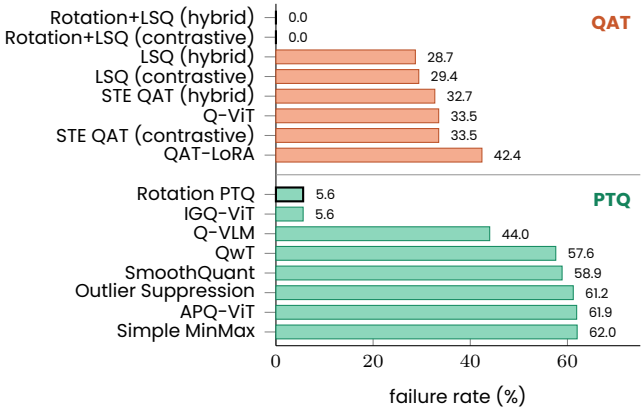
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Means over 10 runs. Quote the table, not Sec. 4.5's prose, which disagrees with it.

- Every PTQ and QAT cell is positive, several times larger at 4 bits. Rotation+LSQ stays near zero at 8 and 6 bits.
- At 4 weight bits every method wins (accuracy kept, ECE improved) in 0% or 2% of runs, against up to 55% at 8 bits (Fig. 28).

Sixteen methods, very different outcomes

A run fails when it loses more than 5% of its zero-shot accuracy

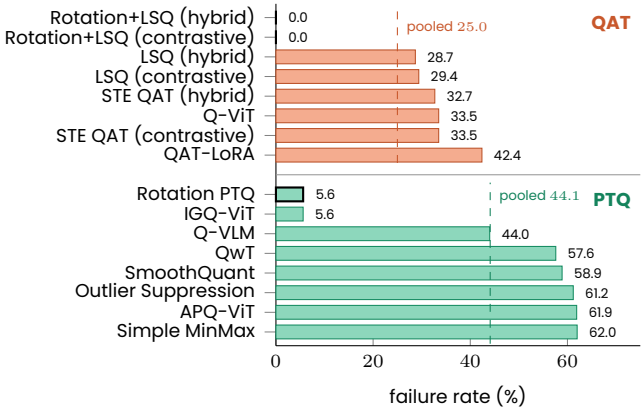


- The rates range from 62.0% for Simple MinMax to 0.0% for Rotation+LSQ.

Transcribed from Bouguerra et al. (2026a, App. F, Figs. 15 and 18).

Sixteen methods, very different outcomes

A run fails when it loses more than 5% of its zero-shot accuracy



- The rates range from 62.0% for Simple MinMax to 0.0% for Rotation+LSQ.
- Pooled, PTQ fails in 44.1% of evaluations and QAT in 25.0%.
- The lowest rates go with rotation, 5.6% for rotation PTQ and 0.0% for Rotation+LSQ, though IGQ-ViT also reaches 5.6%. That rotation is the reason is the paper's reading.

Transcribed from Bouguerra et al. (2026a, App. F, Figs. 15 and 18).

Which method? The case for Rotation+LSQ

The paper's own summary by method family (Table 4, App. B)

at W8A8	PTQ	QAT	**Rot+LSQ**
accuracy	$=/\downarrow$	$=/\uparrow$	$\uparrow$
calibration (ECE)	$\downarrow\downarrow$	$\uparrow$	$\uparrow\uparrow$
OOD detection	$=$	$\uparrow$	$\uparrow$
synthetic-noise robustness	$\uparrow$	$\uparrow$	$\uparrow$
natural covariate shift	$\downarrow$	$\downarrow$	$=$
spurious correlations	$\downarrow\downarrow$	$\downarrow$	$=$
at W4A8	$\downarrow\downarrow$	$\downarrow$	$\downarrow$
failure rate	62.0%	25.0%	0.0%
training cost	none	low	low
hardware kernel support	broad	broad	limited

$\uparrow$ improves, $\downarrow$ degrades, $=$ not significant. PTQ's 62.0% is Simple MinMax; pooled PTQ fails 44.1%. The paper calls natural covariate shift "semantic robustness".

Transcribed from Bouguerra et al. (2026a, Table 4), a qualitative synthesis "only applicable for easy use cases where we observed gains".

- Rotation+LSQ never fails, lowers ECE, and adds no spurious reliance at 8 and 6 bits.
- The paper recommends it for safety-critical settings and names autonomous driving (App. B.1). There are no driving data in the paper.

Which method? The case for Rotation+LSQ

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accuracy	$= / \downarrow$	$= / \uparrow$	$\uparrow$
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natural covariate shift	$\downarrow$	$\downarrow$	$=$
spurious correlations	$\downarrow\downarrow$	$\downarrow$	$=$
at W4A8	$\downarrow\downarrow$	$\downarrow$	$\downarrow$
failure rate	62.0%	25.0%	0.0%
training cost	none	low	low
hardware kernel support	broad	broad	limited

$\uparrow$ improves, $\downarrow$ degrades, $=$ not significant. PTQ's 62.0% is Simple MinMax; pooled PTQ fails 44.1%. The paper calls natural covariate shift "semantic robustness".

Transcribed from Bouguerra et al. (2026a, Table 4), a qualitative synthesis "only applicable for easy use cases where we observed gains".

- Rotation+LSQ never fails, lowers ECE, and adds no spurious reliance at 8 and 6 bits.
- The paper recommends it for safety-critical settings and names autonomous driving (App. B.1). There are no driving data in the paper.
- For a vehicle, the last row matters too. Rotations need kernel support that is still limited.

The deployed engine

CLIP ViT-B/32 (WIT), CIFAR-100, RTX 4070, ten seeds (Sec. 6)

Table 2, clean accuracy and cost

	acc. (%)	latency	speed-up	size
PyTorch FP32	64.47	120.89 ms	1.00×	302.1 MB
TensorRT FP16	64.43	22.70 ms	5.33×	178.7 MB
TensorRT INT8	**66.30**	**15.66 ms**	**7.72×**	**118.4 MB**

INT8 beats FP32 by $+1.83$ points, $p < 10^{-7}$.

Read the baseline. $7.72\times$ is against PyTorch FP32. Against FP16 in the same engine, $1.45\times$ in latency, $1.51\times$ in size, and $+1.87$ points (the notes' arithmetic).

WIT, WebImageText, is CLIP's pretraining set.

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CLIP ViT-B/32 (WIT), CIFAR-100, RTX 4070, ten seeds (Sec. 6)

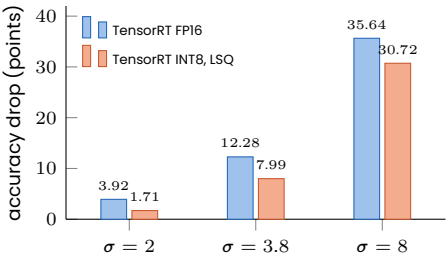
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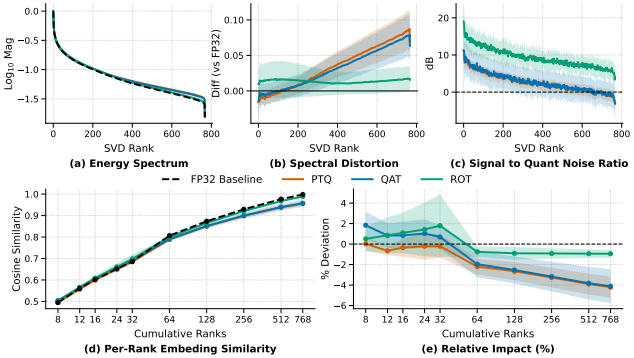
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In Table 3, INT8 loses 2.21, 4.29, 4.92 points less than FP16 under Gaussian blur, all $p < 10^{-8}$. One model, one data set, one GPU.

The VLM study, stage 1, on the fixed FP32 basis

Relative to the signal, the residual weighs most in the tail

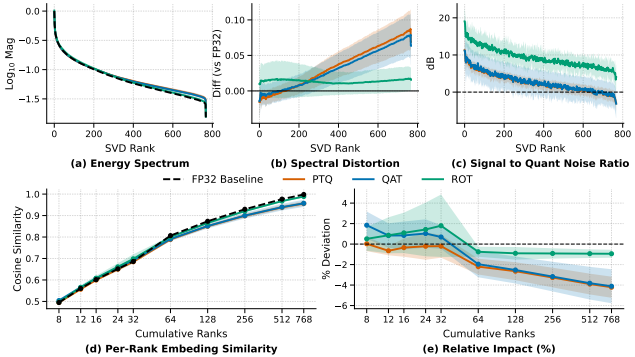


(a) energy spectrum, (b) spectral distortion, (c) per-rank SQNR, (d, e) embedding similarity and impact (Bouguerra et al., 2026a, Fig. 8).

- $SQNR_k$ falls with the rank index for PTQ, QAT, and rotation; rotation keeps it highest. *(measured)*
- "Dampens high-rank components" (Abstract) means a low $SQNR_k$, since the tail's magnitude *rises* in (b).

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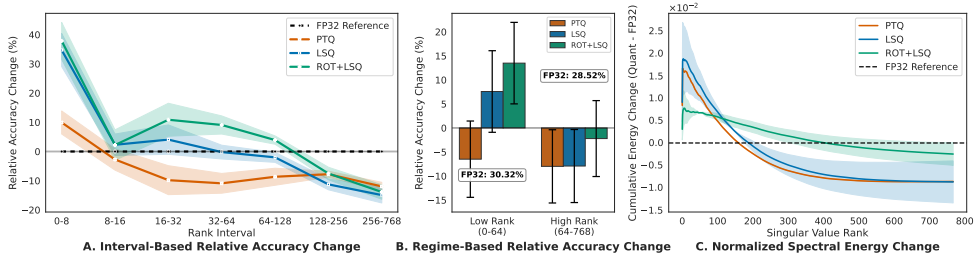


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- "Dampens high-rank components" (Abstract) means a low $SQNR_k$, since the tail's magnitude *rises* in (b).
- That it falls because every direction gets equal error is asserted; isotropy is untested. *(untested)*

The VLM study, stage 2, on the model's own basis

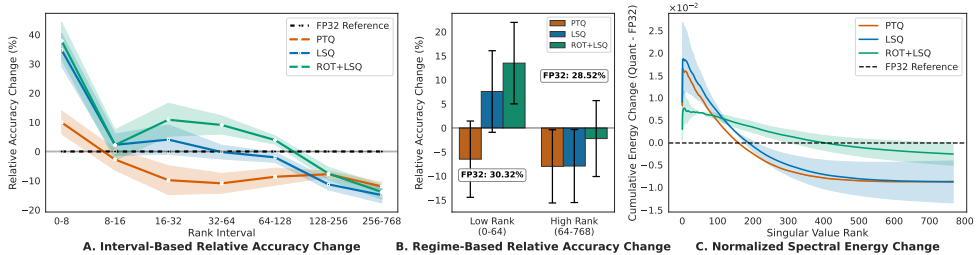
Quantized models lean on the spectral head; the tail still informs



The spectral head band (rank indices 0–8) beats FP32 for PTQ, LSQ and Rotation+LSQ, on one model. The FP32 tail alone (64–768) reaches 28.52%, close to 30.32% for the 64 leading directions, so the tail is information, not noise.

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That tail suppression causes the losses under natural covariate shift and spurious correlation is the paper's reading.

CLIP ViT-B/32 (WIT), ImageNet-1k (Bouguerra et al., 2026a, Fig. 9).

11. Road-scene perception under blur

An overview of the road-scene study; its own talk on day 2 has the details · Notes §18–20

The road-scene study at a glance

DAWN, a classifier, and a matched protocol (Sec. III)

Data

DAWN (Kenk & Hassaballah, 2020), 1026 driving images in real fog, sandstorm, rain, and snow; four classes, car, truck, bus, person.

No model sees a corrupted image in training or range calibration. Gaussian blur only at evaluation, $\sigma = 1$ to 6, probes to 14.

Classifier

Frozen DINOv2 ViT-S/14 with a linear head (the paper's "recognizer"), on 224×224 crops of the annotated objects, so localization plays no part. $N = 1462$ test crops, **balanced accuracy**. W8A8 on the 48 block linear layers, with LSQ activation steps.

The paper also deploys a detector on the same data; its own talk on day 2 covers it.

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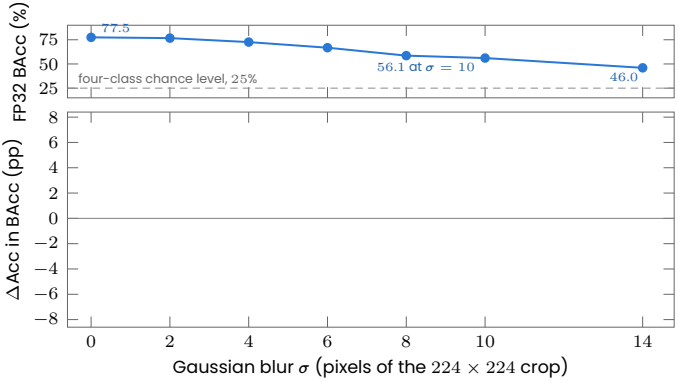
Matched protocol

One clean start. FP32 continues for the same training iterations as QAT, PTQ trains nothing, and a fourth variant trains with matched noise, random noise as large as the rounding error. Same pixels, seeds, and data order. QAT−FP32 measures the whole quantization intervention; which ingredient carries it is untested.

The paper also deploys a detector on the same data; its own talk on day 2 covers it.

The classifier gains under blur, inside a window

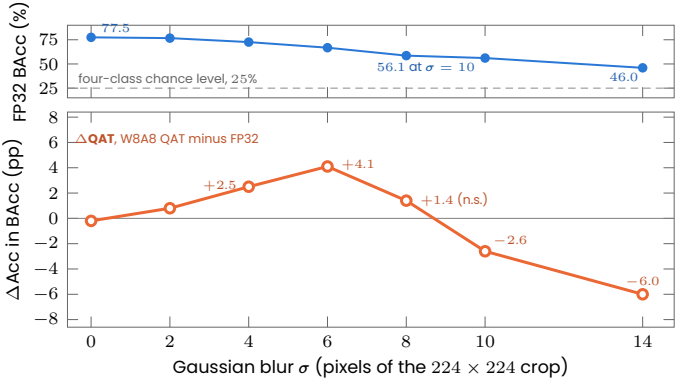
W8A8 QAT against its FP32 control (Tables V, VIII)



Transcribed from Bouguerra et al. (2026b, Tables V, VIII); the paper's Fig. 4 shows the same gain.

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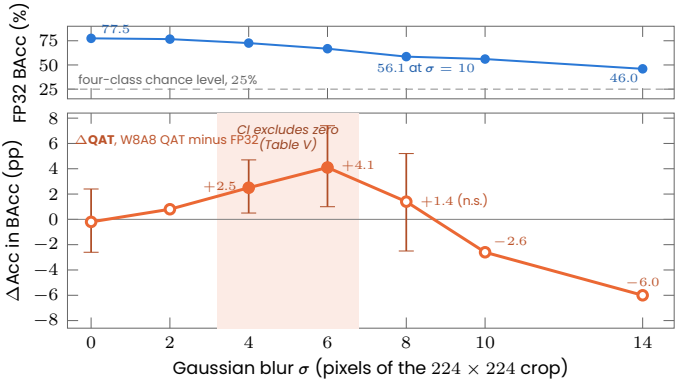


- **A gain from clean training.** $+2.5$ at $\sigma = 4$ and $+4.1$ at $\sigma = 6$; under defocus, $+3.6$ and $+5.9$.

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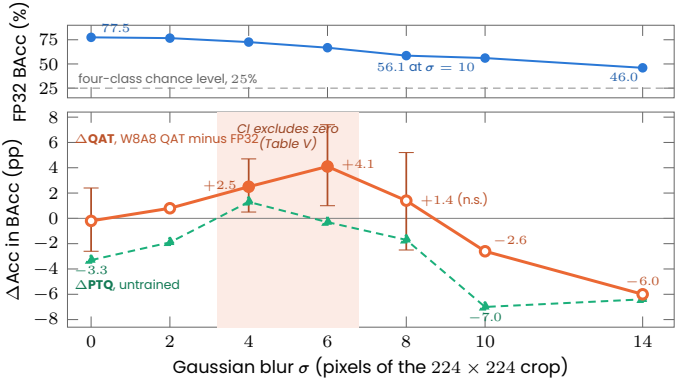


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- Measured against crop sampling on DAWN; only plausible against training variability, since the intervals rest on three runs.

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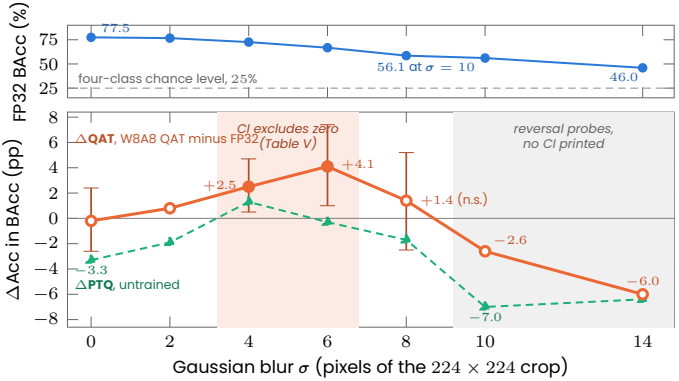


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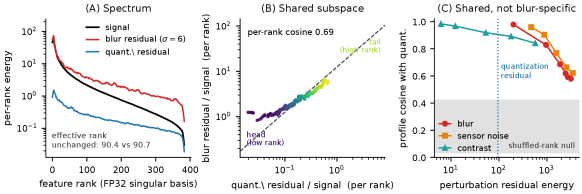


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- PTQ, which trains nothing, shows no such rise under Gaussian blur.
- **Then it reverses**, while FP32 is still 31 points above chance. The cause is untested.

Transcribed from Bouguerra et al. (2026b, Tables V, VIII); the paper's Fig. 4 shows the same gain.

Why it might work, and what is still open

A shared feature tail, and a noise-size control (Sec. VII)



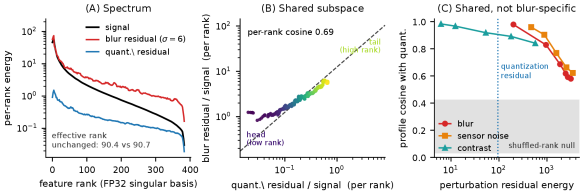
(A) Residual energy per rank. (B) Blur against quantization, rank by rank. (C) The match for three corruptions.

The same weak directions.
Rounding and blur change the features most in the tail, and alike rank by rank (cosine 0.69). *(measured)*
In the paper's reading, a model trained to tolerate its own rounding error sees blur in directions it already tolerates. *(plausible)*

Bouguerra et al. (2026b, Fig. 6). Limits (Sec. VIII): one benchmark, encoder, and laptop GPU; blur on native weather.

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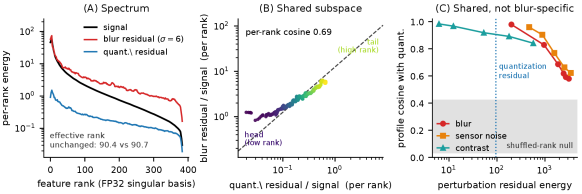
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Not specific to rounding.
Sensor noise and lower contrast hit the same directions just as well (C). So the overlap may be needed, but it cannot explain the gain alone.

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Why it might work, and what is still open

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Not specific to rounding. Sensor noise and lower contrast hit the same directions just as well (C). So the overlap may be needed, but it cannot explain the gain alone.

Noise of the same size is not enough. A model trained with random noise as large as its rounding error gains on clean crops ($+1.8$), not under blur, where QAT gains. *(plausible)* What does the work is untested.

Bouguerra et al. (2026b, Fig. 6). Limits (Sec. VIII): one benchmark, encoder, and laptop GPU; blur on native weather.

12. What it means for automotive perception

What is measured, what to do, and what is open · Notes §21

What is measured, plausible, and untested

The claims of both studies, the VLM study first

Measured

- Not systematically destructive; 7.8% of W8A8 runs improve accuracy and calibration.
- Robustness rises under synthetic corruption ($+8.9$), falls under natural shift (-0.3).
- Rotation+LSQ never fails, and adds no spurious reliance at 8 and 6 bits.
- A road-scene classifier gains $+2.5$ to $+5.9$ inside a blur window, then reverses.

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- Rotation+LSQ suits driving, the VLM study's advice; it has no driving data.
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Untested

- The shared tail is necessary for the gain.
- Clipping at the learned step does the work.
- Coarser activations widen the window.
- Why the gain reverses while FP32 still scores 56.1 at $\sigma = 10$.

Practices for a perception team

What the two studies support today

1. **Evaluate per severity, per corruption family, and per class, on driving data.** A suite average dilutes a window, and plain accuracy hides a minority-class gain.

Practices for a perception team

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3. **Recalibrate confidence after quantizing.** One refitted temperature leaves every prediction unchanged, and score thresholds depend on it. Fit it on data like the deployment's; at a 0.8 threshold an overconfident model can promise 91% and deliver 78% (synthetic).

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4. **Prefer a trained quantizer to naive PTQ.** LSQ on clean data "needs no corrupted data, no extra loss term, and no architectural change" (road-scene study); Rotation+LSQ never failed in the VLM study's grid, if the target supports rotations.

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Below 8 bits, test again

The evidence stops at 8 bits. At W4A8 even Rotation+LSQ leans more on spurious context, and the blur gain was never measured there. And the gain is one benchmark wide, so treat it as a hypothesis to verify on your own data and hardware.

Open problems, each an experiment

Change one thing at a time, three seeds, the same FP32 control

What in the quantizer does the work?

1. **Clipping without rounding**, at the step QAT learned. If the window returns, clipping carries it.
2. **Rounding without clipping.** If neither 1 nor 2 reproduces the gain, their interaction does.
3. **A fairer noise control**, with noise that follows the learned step, as many trained numbers as QAT, and noise fixed per input against noise drawn anew.
4. **The bit-width sweep**, activations at 8, 6, 5, 4 bits. The road-scene account predicts a wider window (the paper's limitation 2).

Does it hold on the road?

5. **A second dataset and encoder family**, and a model trained on blurred images as the best case, the paper's own future work.
6. **Natural weather alone**, against synthetic blur on top of it (the paper's limitation 4).
7. **The classifier on detected boxes**, rather than on crops of annotated objects.

Start in the notebook

Part 5 runs the classifier protocol at one seed; its exercise (c) pilots experiment 4.

One more problem: flips under matched accuracy

A teaser for the group's NeurIPS 2026 paper

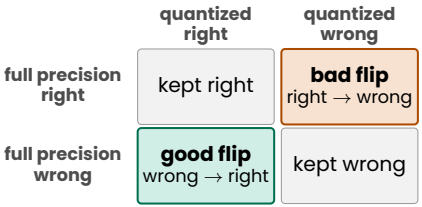
	quantized right	quantized wrong
full precision right	kept right	**bad flip** right → wrong
full precision wrong	**good flip** wrong → right	kept wrong

- **Matched accuracy hides flips.** A quantized model that matches its full-precision baseline still changes many predictions, good ones and bad ones, in roughly equal numbers.
- **They are not random.** Good flips peak at low ranks of the quantized model's SVD spectrum (median 24), bad flips in the tail (median 48), over $13{,}205$ flips of CLIP-B/16 at W8A8 QAT.

Bouguerra et al. (2026c), paper coming soon; project page and code online.

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- **They are not random.** Good flips peak at low ranks of the quantized model's SVD spectrum (median 24), bad flips in the tail (median 48), over $13{,}205$ flips of CLIP-B/16 at W8A8 QAT.
- **A cheap repair.** A gate, a small model fitted afterward with labels, reads the spectrum, keeps the good flips, and repairs the bad ones. It gains $+1.97$ points over the quantized model and $+0.86$ over full precision across the paper's 50 CLIP settings, at 0.045% of the compute.
  The gate needs labels for each model pair, and the gains are upper estimates.

Bouguerra et al. (2026c), paper coming soon; project page and code online.

The sentence to leave with

Quantized to 8 bits, and most dependably when the quantizer is trained, a perception model can become more reliable in specific, measurable places and less reliable in others.

Both papers see its residual rise, relative to the signal, toward the tail of the feature spectrum. In the road-scene study, noise matched to the step does not stand in for it. **Neither has isolated what in the quantizer does the work.**

Questions?

Happy to discuss quantizers, reliability, and the two papers

THE TWO PAPERS

Bouguerra et al. (2026a), ICML 2026, arXiv:2509.21173

Bouguerra et al. (2026b), IEEE ICVES 2026, more details in its presentation on day 2

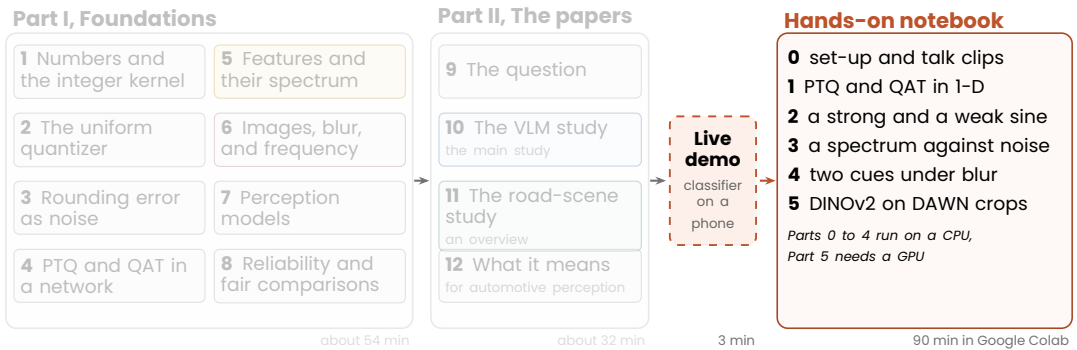
CONTACT

fabio.arnez@cea.fr

The companion lecture notes hold every derivation, figure, and number with its locator.

Next, the hands-on session

The same argument, with every quantity on a dial



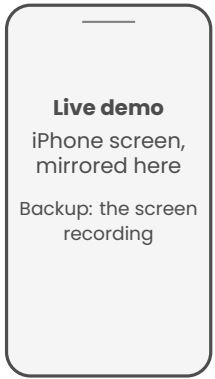
Parts 1 to 4 make the argument adjustable: bit width, feature strength, and corruption level. Part 5 re-runs the recipe on the real road-object encoder.

Two cautions. Part 5 gains move 1 to 3 points with the seed, run, and hardware. The toy parts explain robustness by noise size alone; read Exercise 3.2 against the matched-noise control.

The classifier, live on a phone

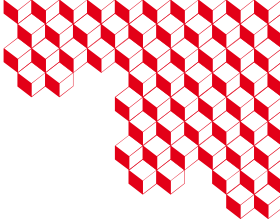
Our re-run of the road-scene recipe, on an iPhone

- Road objects in adverse weather, DAWN crops of cars, trucks, buses, and people.
- One model, DINOv2 ViT-S/14 with a linear head, in three precisions:
  - FP16, the full-precision control;
  - PTQ W8A8, quantized after training;
  - QAT W8A8, trained with the 8-bit quantizer in the loop.
- All three run live and offline on the Neural Engine. Training saw clean images only; blur comes at test time.
- The 8-bit weights take half the memory, 22.1 against 43.3 MB.
- At $\sigma = 6$, **QAT beats FP16 by 5.6 points** of balanced accuracy on the phone and by 5.8 in simulation, over three seeds (the paper: $+4.1$ against FP32).



iPhone 15 Pro, app “Less Precise”

1462 test crops, three seeds; bit for bit our study run on this phone. Recipe: Bouguerra et al. (2026b).



Thank you !

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